

Slings: Autologous, Biologic, Synthetic, and Mid-Urethral

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PATHOPHYSIOLOGY OF STRESS URINARY INCONTINENCE AND SLING MECHANISM OF ACTION

Two theories dominate the explanation of how a sling works.

DeLancey (1994) — the Hammock Hypothesis

- The healthy female urethra rests on a **"suburethral hammock"** attached laterally to the **arcus tendineus fasciae pelvis (ATFP)**.
- In stress urinary incontinence (SUI) this supporting layer is deficient, **manifesting clinically as hypermobility of the bladder neck and urethra**.
- Inferior descent of these structures produces **unequal transmission of intra-abdominal pressure**, and hence leakage.
- Therapeutic corollary: a **pubovaginal sling (PVS) sited at the bladder neck** abolishes SUI by supplying a strong tissue layer that gives **dynamic compression of the urethra during rises in intra-abdominal pressure**.

The Mid-Urethral Theory

- Ingelman-Sundberg (1953): the **pubococcygeus muscles insert at the level of the mid-urethra**, just outside the vaginal epithelial wall, and are integral to the mid-urethral continence mechanism.
- Asmussen and Ulmsten (1983), Westby (1982): **maximum urethral closure pressure (MUCP) occurs at the mid-urethra** in continent females — attributed to the combined contribution of the anatomical structures sited there.

Petros and Ulmsten (1990) — the Integral Theory

- A unifying concept: continence depends on adequate function of **three components — the pubourethral ligaments, the suburethral vaginal hammock, and the pubococcygeus muscles**.
- Injury to **any one** of the three — by **surgery, parturition, ageing, or hormonal deprivation** — impairs mid-urethral function and produces **urinary incontinence (UI)**.
- Supporting evidence (Ulmsten et al., 1987): skin and round ligament biopsies showed **incontinent females had substantially less collagen — classically quoted as 40% less — than continent controls**, the inference being that **collagen loss weakens the connective tissue supporting the urethra**.

THE ADVENT OF THE MID-URETHRAL SLING

- **Ulmsten and Petros (1995)** applied the integral theory to build the **first retropubic synthetic mid-urethral sling (MUS)**, named **intravaginal slingplasty**; the initial report gave good cure rates with no reported complications.
- **Ulmsten et al. (1998)** renamed the procedure the **tension-free vaginal tape (TVT)** and reported high cure rates in a multicentre study.
- Complications in that TVT series established the **characteristic complication profile**: transient urinary retention, one case requiring a small adjustment of the sling through a vaginal incision, haematomas, and **mesh perforation of the bladder recognised intraoperatively**; vaginal mesh exposure could not be

corroborated, and one further patient had a wound infection needing a minor procedure and vaginal oestrogen.

- **This initial trial was the first to report the two complications unique to and defining of MUS surgery: mesh penetration of the urinary tract, and vaginal mesh exposure.**
- **Delorme (2001)** described the **transobturator approach** specifically **to minimise perforation of the urinary tract, bowel and neurovascular structures**; his initial report showed high cure with a single complication.
- **Richter et al. (2010) — the TOMUS trial**, ten years after transobturator MUS was introduced: **retropubic and transobturator MUS are similarly efficacious and generally safe.**
- **2006: the United States Food and Drug Administration (FDA) approved the first single-incision MUS (SIMS) — performed entirely vaginally, with no skin entry.**
- Sling procedures of all types have been **consistently shown to be efficacious and safe** for female SUI; over the same period **regulatory and professional bodies have raised concerns about mesh exposure and perforation** specific to synthetic mesh, biologic grafts, and the specialised placement tools. Detailed discussion is in Chapter 119.

Guideline Positions

- **American Urological Association / Society for Urodynamics and Female Urology (AUA/SUFU) Guideline on the Surgical Treatment of Female SUI** (Kobashi et al., 2017, 2023):
- **Strong Recommendation** — physicians may offer MUS to **index patients** considering SUI surgery.
- **Conditional Recommendation** — **either retropubic or transobturator MUS** may be offered to index patients electing surgery.
- **Both rest on grade A evidence.**
- SIMS outcome data accumulating over recent years produced a **Conditional Recommendation in index patients on grade B evidence** in the **2023 AUA/SUFU Update.**

KEY POINTS: INTRODUCTION TO SLINGS

- **Early theories of SUI centred on unequal pressure transmission to bladder and urethra, urethral hypermobility, and an excessively caudal urethral position.**
- **Later theories emphasised the pubourethral ligaments, suburethral vaginal hammock and pubococcygeus muscle** as the vital supports of the proximal and mid-urethra.
- **Consequent surgical principle: PVS are placed under minimal tension at the bladder neck to re-establish the suburethral hammock; MUS are placed loosely under the mid-urethra to prevent movement of the posterior urethral wall and assist urethral coaptation.**

PREOPERATIVE ASSESSMENT

The purpose of evaluating the incontinent female is to **characterise the type of incontinence**, assess the contribution of **comorbid medical conditions**, elucidate the **differential diagnoses**, and obtain **prognostic information guiding treatment selection**.

The Initial Evaluation

AUA/SUFU Clinical Principle — five components in a female with SUI desiring surgery

- **Focused history, including assessment of bother** (e.g. validated questionnaire)
- **Focused physical examination, including pelvic examination**

- **Objective demonstration of SUI with a comfortably full bladder** — Valsalva or **cough stress test (CST)**
- **Assessment of post-void residual urine volume (PVR)** — any method acceptable
- **Urinalysis**

The **European Association of Urology (EAU)** has separately published guidelines on **assessment**, on **non-surgical** treatment (Nambiar et al., 2018) and on **surgical** treatment of urinary incontinence (Lucas et al., 2013).

History

- Define the **onset, frequency, character and severity** of the incontinence, plus other **storage and voiding symptoms**.
- Identify **aggravating factors** — notably **sexual intercourse and exercise**.
- **Accurately assess baseline urinary urgency: this specific symptom correlates with poorer outcomes after sling surgery** (Richter et al., 2008).
- Cover factors affecting bladder and urethral function: **neurological disease, prior surgery, pelvic radiation, current medications**.
- **Ask specifically about faecal incontinence and defaecatory problems**.
- Because incontinence **affects quality of life (QoL) rather than quantity of life, formal assessment of bother is integral**, not optional.

Examination

- A **focused neuromuscular and pelvic examination is mandatory** in any female complaining of urinary incontinence.
- It may uncover **neurological or musculoskeletal disorder**, and **correctable anatomical abnormalities** such as **vesicovaginal fistula or vaginal epithelial irritation**.
- Standard components: **vaginal anatomy** — urethral meatus and vaginal support, i.e. **pelvic organ prolapse (POP)**; **quality of vaginal tissue** — atrophy, irritation;

degree of urethrovesical hypermobility; associated findings — fistulae, foreign material.

- **Valsalva during examination to reveal or exaggerate POP, urethral hypermobility and SUI**.
- **Cotton-tipped applicator (Q-tip) test, Bergman and Bhatia (1987): hypermobility is defined as movement >30 degrees from baseline on straining** — though the test's usefulness is **controversial**.
- **If a supine stress test with a full bladder is negative, repeat it standing**.

Guideline Positions on Further Testing

Baseline

- **Urinalysis and PVR in all females; more extensive imaging is not part of the initial evaluation of SUI**.

Additional evaluation SHOULD be performed (Expert Opinion)

- **Inability to make a definitive diagnosis** from symptoms and initial evaluation
- **Inability to objectively demonstrate SUI**
- **Known or suspected neurogenic lower urinary tract (LUT) dysfunction** — e.g. multiple sclerosis
- **Abnormal urinalysis** — unexplained haematuria or pyuria
- **Urgency-predominant mixed urinary incontinence (MUI)**
- **Elevated PVR by clinician judgement**
- **High-grade POP** — pelvic organ prolapse quantification (POP-Q) stage ≥ 3 — with SUI not demonstrated on POP reduction
- **Significant voiding dysfunction**

Additional evaluation MAY be performed

- Concomitant **overactive bladder (OAB)** symptoms; **failed previous anti-incontinence surgery**; **previous POP surgery**.

Modalities of additional testing

- **Upper tract radiography** where there is **concomitant haematuria or suspected neurogenic dysfunction**.
- **Magnetic resonance imaging (MRI)** of head, spine and pelvis to detect **spinal cord or brain lesions**.
- **Cystoscopy** should **NOT** be performed in evaluating SUI in the index patient (Clinical Principle). It becomes integral, however, where there is concern for neoplasia or foreign body, or after previous pelvic surgery.

The Urodynamics Controversy

- Preoperative urodynamic evaluation — **filling cystometrogram, leak point pressures, electromyography, pressure-flow studies** — remains an area of controversy.
- **AUA/SUFU Conditional Recommendation, grade B evidence: urodynamics may be OMITTED in the index patient when SUI is clearly demonstrated.**

Landmark Trial — ValUE (Nager et al., 2012)

- **Multicentre randomised non-inferiority trial, 630 females** with SUI or stress-predominant MUI, randomised to **urodynamics versus office evaluation alone** before anti-incontinence surgery.
- **Conclusion: preoperative office evaluation alone was non-inferior to evaluation with urodynamics for one-year outcomes in uncomplicated, demonstrable SUI.**
- Secondary analysis (Norton et al., 2016): urodynamics carried **substantial per-patient cost, in excess of US\$300, extrapolating to tens of millions of dollars of annual United States expenditure** — cited alongside the **invasiveness of the studies** as further argument against routine use.

Criticism of the trial's design

- **The overwhelming majority of screened females were ineligible** for inclusion — taken to indicate that **most females presenting with SUI do NOT have uncomplicated SUI**, so the trial population is unrepresentative of practice.
- **Almost all treating physicians were fellowship-trained**, arguably reducing the need for rigorous urodynamic testing in that setting and **limiting generalisability to non-specialist practice**.

The Counter-Argument: Assessment of Urethral Function

- The chief goal urodynamics uniquely serves is **evaluation of urethral function**, indexed by **low MUCP** and by **abdominal leak point pressure (ALPP) / Valsalva leak point pressure (VLPP)**.
- **McGuire (1981): MUCP <20 cm H₂O indicates type 3 SUI** — now termed **intrinsic sphincter deficiency (ISD)**.
- **Sand et al. (1987)**, in females undergoing modified Burch colposuspension: **MUCP ≤20 cm H₂O predicted a markedly higher failure rate than MUCP >20 cm H₂O**.
- **MUCP ≤20 cm H₂O** has since been used generally to characterise severe urethral dysfunction.
- **McGuire et al. (1993)** — the **ALPP thresholds: ALPP <60 cm H₂O signifies ISD; ALPP >90 cm H₂O signifies little or no ISD**.
- The definition of ISD has since evolved from strict urodynamic criteria to a clinical diagnosis.
- **Conceptually essential: every female with stress incontinence has some component of ISD**. ISD is not a separate disease entity but a matter of degree.

- **Urodynamics remain the only way to definitively diagnose ISD and low MUCP/ ALPP/VLPP preoperatively.**

Where the evidence sits

- **No study has shown that urodynamic evaluation definitively predicts postoperative outcome in uncomplicated SUI.**
- Multiple authors report that **low MUCP or VLPP predicts poorer outcome after MUS, and specifically after the transobturator approach.**
- Others found **no predictive value** of low MUCP or VLPP after MUS (Costantini et al., 2008, 2009).
- Further refinement proposed: **only ISD in the presence of a hypomobile urethra portends poorer success** (Wlazlak et al., 2017) — ISD remains an **incompletely understood concept.**

Urodynamics in the Non-Index Patient

Expert Opinion — urodynamics MAY be performed

- **Prior anti-incontinence or POP surgery**
- **Mismatch between subjective and objective measures**
- **Significant voiding dysfunction**
- **Urge-predominant MUI, urgency urinary incontinence (UUI), or OAB**
- **Elevated PVR**
- **Unconfirmed SUI**
- **Neurogenic LUT dysfunction**
- The decision turns on the **patient's willingness to undergo the study** and on **whether the result would alter or better define treatment options.**
- **In significant POP, urodynamics must be performed both with and without POP reduction — using a sponge stick, speculum or pessary.**

Bladder-side information from urodynamics

- Beyond outlet assessment, **filling and emptying pressures give prognostic information about bladder function.**
- **Detrusor underactivity and Valsalva voiding may predict postoperative voiding difficulty** (Gracely et al., 2021) — **more prominently in non-index patients.**
- **Detrusor overactivity (DO) found on filling may change the choice of treatment — the single most consequential bladder finding.**

The two-way relationship between SUI and detrusor overactivity

- Mechanism proposed for **stress-induced DO: stress leakage of urine into the proximal urethra** (Kuru, 1965) and **traction on pelvic nerves when raised abdominal pressure acts on weakened pelvic supportive tissue** (Serels et al., 2000).
- Therefore **pre-existing DO may IMPROVE after anti-incontinence surgery** where the DO is stress-induced.
- **But existing DO and urgency may equally WORSEN** after otherwise uncomplicated anti-incontinence surgery.
- Practical consequence: **in known DO, consider alternative or complementary treatments before proceeding to sling surgery.**
- **Low cystometric bladder capacity and decreased compliance may also worsen sling outcomes** and must be weighed.
- **The overall value of urodynamics in predicting sling outcomes remains debatable** (Nager et al., 2008, 2011).

KEY POINTS: PREOPERATIVE ASSESSMENT

- **Minimum evaluation** of a female with UI: **focused history, physical examination, and objective confirmation of SUI**, plus the **basic clinical tests — urinalysis and PVR**.
- **Cystoscopy should not be performed in uncomplicated SUI**, but **cystoscopy and radiographic imaging may be performed in haematuria, prior pelvic surgery, or suspected urinary tract abnormality**.
- **Urodynamic studies (UDS) may be omitted in uncomplicated SUI and stress-predominant MUI desiring surgery**; they prove useful where **the diagnosis is unclear** or where **complicating factors** coexist — significant storage and voiding symptoms, prior LUT surgery, suspected neurogenic bladder dysfunction, or significant POP.
- Slings treat **both SUI and stress-predominant MUI effectively**, but **thorough characterisation of preoperative symptoms and bother guides counselling and treatment choice**.

ALTERNATIVE TREATMENT OPTIONS

- Because UI rarely affects quantity of life, **non-surgical options must be discussed with every patient**.
- **AUA/SUFU Expert Opinion: treatment should be tailored to the individual, and patient bother taken into account in deciding on therapy**.

Initial Non-Surgical (Conservative) Therapy

- **Patient education • behavioural and dietary modification • judicious fluid restriction • weight loss • pelvic floor muscle training (PFMT)**.

Weight loss

- **Subak et al. (2009a)**: randomising females with obesity between two weight-loss programmes, **greater weight loss produced correspondingly greater reduction in SUI episodes — a dose-response relationship** between weight lost and continence gained.
- A recent review concluded **weight loss benefits both SUI and OAB symptoms** (Fuselier et al., 2018).

Pelvic floor muscle training

- **Kegel exercises and formal PFMT** act by improving **pelvic floor muscle strength and coordination**.
- **Cochrane systematic review (Dumoulin et al., 2018)**, pooling randomised and quasi-randomised trials of PFMT in female SUI: **PFMT should be recommended to most patients BEFORE surgical treatment**. Versus no treatment, PFMT gave **markedly higher rates of cure or improvement of SUI**, and roughly **double the rate of cure or improvement of incontinence of any type**.
- **Adherence is central** to both short- and long-term symptom improvement — the limiting factor in practice.

Devices

- **Pessaries and other vaginal and urethral devices help SUI in selected individuals**.

Periurethral bulking agents

- **Submucosal injection via cystoscope — a minimally invasive option with variable success and frequent need for retreatment**.
- **Cochrane review (Kirchin et al., 2017): limited evidence of benefit**.
- Nonetheless **remains an option for females not ready for a more morbid and invasive procedure**. Detailed in Chapter 120.

PUBOVAGINAL SLING

Defining contrast: the PVS is placed under the **BLADDER NECK**; the MUS is placed under the **MID-URETHRA**.

Indications — the versatility of the PVS

- **Primary and recurrent SUI**, associated with **both urethral hypermobility AND ISD**.
- **Pure SUI and stress-predominant MUI**.
- **Concomitant antero-apical POP**.
- **Complex SUI associated with neurological conditions**.
- **Neurogenic SUI**, e.g. **myelodysplasia** — effective for **leakage occurring between clean intermittent catheterisations (CIC)**, but **only once urodynamics have confirmed sufficient bladder capacity and compliance**.
- **Urethral reconstruction** after damage from **trauma, synthetic mesh or graft perforation**, and **iatrogenic hypospadias**.
- **Interposition during repair of urethrovaginal fistula or urethral diverticulum**.
- **Recurrent SUI after failed retropubic suspension or failed MUS**.

Of the many materials and techniques described, the **PVS using AUTOLOGOUS FASCIA** remains the **gold standard for management of ALL forms of SUI**.

Anatomy and Mechanics of a Pubovaginal Sling

Mechanism of stress leakage

- The female urethra **lies beneath the pubic symphysis**; the **pubourethral ligaments** suspend the **anterior urethral wall to the pubic arch**.
- In **urethral hypermobility**, a rise in intra-abdominal pressure makes the **posterior urethral wall slide away from the anterior wall**, thereby **opening the bladder neck and proximal urethra**.
- **Uneven pressure transmission plus bladder neck opening (funnelling)** produces the stress leak.
- **ISD is a different defect entirely** — it arises **within the urethra proper**, so the sphincter **cannot coapt or generate sufficient resting urethral closing pressure**.

The four-layer urethra and the components of continence

- The female urethra comprises **four separate tissue layers** that keep it closed. The **middle muscular layer** compresses to maintain the resting urethral closure mechanism; the **outer seromuscular layer** augments that closing pressure.
- **Normally the resting urethral closing pressure of the internal sphincter exceeds the resting or Valsalva pressure of the bladder**.
- External sphincter fibre types: **fast-twitch fibres → sudden voluntary contraction**; **slow-twitch fibres → continuous passive control via the involuntary guarding reflex during filling**.
- **Levator ani** maintains the integrity of the pelvic diaphragm for continence control.
- Support structures: **urethropelvic ligament → supports the bladder neck**; **pubocervical fascia → supports the undersurface of the bladder**.

How the sling works

- The bladder neck PVS gives **dynamic urethral compression WITHOUT obstruction** during rises in intra-abdominal pressure.
- **Goal: adequate urethral coaptation plus increased urethral responsiveness to abdominal pressure** — balanced against **ischaemia, retention and erosion from unnecessary tension**.

- **Blaivas and Olsson (1988)** described the current version of the PVS: a **SHORT FREE GRAFT** of rectus fascia whose tension could be adjusted.
- **Critical mechanistic point:** it is incorporation of the sling into the endopelvic fascia, with subsequent fixation and fibrosis — **NOT** entry into the retropubic space — that prevents the sling giving way under raised intra-abdominal pressure.
- **Sling width <2 cm:** wide enough to give the necessary urethral compression, with **cross-sectional area** sufficient to avoid a narrow constricting band.

Pubovaginal Sling Materials

The ideal and the reality

- Materials used across the last century fall into four classes: **autologous, allograft, xenograft, synthetic**.
- The **ideal material** gives **long-lasting suburethral support** with **minimal adverse effects**, and **incorporates into the host** with **minimal tissue reaction**.
- In reality, most materials work by **promoting organised fibrosis**, reinforcing the sphincteric mechanism through improved suburethral support.
- The **fibrosis paradox:** theoretically **greater fibrosis → better clinical results**, but **inflammatory infiltration** causes **rapid sling degradation** and **possible tissue destruction with erosion**.
- **Autologous slings** are **completely biocompatible** and **rarely associated with urethral perforation**.
Biologic grafts and synthetics gained past popularity only for **reduced operative time, morbidity, pain and hospital stay**.
- Most allograft, xenograft and synthetic sling materials are **no longer available**; further discussion of synthetics for **bladder neck sling (BNS)** construction is in Chapter 119.

Autologous Pubovaginal Slings

The two commonest autologous materials

- **Rectus abdominis fascia** — harvested from the abdominal wall.
- **Fascia lata** — harvested from the lateral thigh.

Histological behaviour

- **FitzGerald et al. (2000):** rectus fascial grafts undergo **extensive remodelling after placement**, with **abundant fibroblasts and connective tissue** on biopsy.
- **Woodruff et al. (2008)**, comparing synthetic, autologous, allograft and xenograft slings histologically: **autologous materials showed the greatest host fibroblast infiltration and neovascularisation**, with **minimal inflammatory or foreign-body reaction**.
- Explants examined up to 65 months after placement showed **autologous fascial grafts consistently intact** with **minimal gross degradation**.
- **Autologous tissue** carries a **negligible risk of urethral erosion** (Webster and Gerridzen, 2003).
- **Disadvantages of harvest:** **increased operative time, longer hospital stay, more postoperative pain, and harvest-site complications**.

Rectus fascia — the practical point

- **Even where the harvest site is scarred and thickened by prior operations**, graft quality is **not necessarily compromised**.
- **Prior ventral hernia repair** is the situation in which **fascia lata** becomes the preferred autologous material.

Fascia lata — the comparison with rectus fascia

- **Similar:** **completely biocompatible, minimal tissue reaction, similar mechanical properties**.

- **Better:** shorter reported recovery time; no risk of abdominal harvest-site complications.
- **Worse:** requires patient repositioning, additional operative time, and operating in an area unfamiliar to most pelvic surgeons; **thigh pain is a short-term sequela.**
- **Harvest technique matters for thigh pain:** early series reported **pain on walking in the majority at one week**, whereas harvest with a **Crawford fascial stripper** left **only a small minority with incision-site pain at one week** (Latini et al., 2004).
- **Thigh muscle herniation is width-dependent:** a 10–20 cm wide graft produced herniation in roughly half; a narrow graft, 1.5 cm × 12–15 cm, produced NONE.

Vaginal epithelium (vaginal wall sling)

- **Raz et al. (1989)** described the use of **in situ vaginal wall**.
- **Drawbacks:** possibly insufficient tensile strength; risk of epithelial inclusion cyst formation; vaginal shortening.
- **The absence of retropubic space dissection may mitigate the overall efficacy of the procedure.**

KEY POINTS: PUBOVAGINAL SLING MATERIALS

- The ideal PVS material is sterile, biocompatible, non-carcinogenic and consistent in quality.
- The implanted material should **incorporate into the host with minimal tissue reaction**, giving **long-lasting suburethral support with minimal complications**.
- **Autologous materials remain the gold standard for PVS — no adverse tissue reaction and negligible chance of urethral perforation.** Other biomaterials were adopted only to **shorten operative time, hospital stay and recovery**.
- **Allografts:** tissue-processing techniques vary and may disrupt the microstructure, affecting ultimate graft strength.
- **Xenografts:** in situ, less tensile strength than allografts, and the highest propensity to encapsulate; processing likewise varies significantly.
- **Synthetics:** significant inflammatory and foreign-body reaction, with higher rates of graft infection and perforation — **SUBOPTIMAL** for placement at the bladder neck.

AUTOLOGOUS PUBOVAGINAL SLING OPERATIVE PROCEDURE

Counselling

- Cover the **global risks of surgery** in informed consent, then the sling-specific risks.
- **Counsel explicitly on transient AND permanent voiding dysfunction:** postoperative difficulty emptying the bladder, and **de novo urgency and frequency**.
- **Preoperative instruction in CIC** in selected patients may avoid an indwelling urethral catheter postoperatively.
- **Uberoi et al. (2021) — expectation management is a distinct counselling task:** focus-group participants expected sling surgery to resolve other pelvic floor issues such as bowel symptoms, and **described persisting non-SUI symptoms as negative outcomes even when their SUI had improved.**

Anaesthesia, Patient Positioning and Preparation

Anaesthesia

- **Spinal or general — decided by patient, surgeon and anaesthetist consensus.**

Antibiotic prophylaxis

- **Sling surgery is a CLEAN-CONTAMINATED case.**
- Prophylaxis must cover *Staphylococcus aureus*, streptococci, enterococci, vaginal anaerobes and skin flora.
- **AUA Best Practice Policy Statement:** a single preoperative dose of a first- or second-generation cephalosporin. Second-generation agents (cefoxitin, cefotetan) give better anaerobic cover in principle than first-generation cefazolin — however, cefazolin proved equivalent for vaginal anaerobes in sling procedures (Lightner et al., 2020).
- **Alternative single-dose regimens:** ampicillin/sulbactam · aminoglycoside or aztreonam with metronidazole · aztreonam with clindamycin · clindamycin alone.
- **Systematic review (Temtanakitpaisan et al., 2022):** prophylactic antibiotics had little or no effect on surgical site infection, urinary tract infection (UTI), bacteriuria, postoperative fever, or vaginal mesh exposure — but the included trials were small and the conclusion must be interpreted with caution.

Thromboprophylaxis

- **Bilateral lower-limb intermittent pneumatic compression (IPC) devices routinely applied BEFORE induction of anaesthesia.**
- **AUA Best Practice Policy Statement for Prevention of Deep Venous Thrombosis in Urologic Surgery (Forrest et al., 2009):** for incontinence surgery, the choice between IPC, low-dose unfractionated heparin and low-molecular-weight heparin is made on individual patient and procedural risk factors — no blanket recommendation.

Positioning and nerve injury

- **Dorsal lithotomy**, with deliberate care to **minimise lower-limb neural injury.**
- **Bohrer et al. (2009):** overall incidence of peripheral nerve injury is low. **Commonest nerves injured:** lateral femoral cutaneous and femoral (equal, and jointly predominant), then common fibular (common peroneal), ilioinguinal/ iliohypogastric, saphenous and genitofemoral.
- **Prognosis is excellent:** complete resolution of neuropathic symptoms in all but one patient, at a median of approximately one month (range one day to six months).

Skin and vaginal preparation

- **Prepare and drape the abdomen from the umbilicus down, and the vagina.**
- **Povidone-iodine is the commonest vaginal antiseptic in the United States;** practice differs elsewhere.
- **American College of Obstetricians and Gynecologists (ACOG) committee opinion, September 2013:** chlorhexidine gluconate solutions containing 4% alcohol or less are safe and effective for vaginal preparation — an off-label use.

Set-up

- **Weighted speculum** in the vagina; **18-Fr Foley catheter** per urethra.
- **Moderate Trendelenburg;** a **headlight** aids vaginal dissection.
- A **vaginal ring retractor** retracts the **labia majora**, and later the **incision**, improving exposure.

Additional set-up for fascia lata harvest

- **IPC device positioned BELOW the patella on the harvest side.**
- **Knee elevated and supported** on a 1-litre fluid bag or cushion.
- **Limb internally rotated at the hip and taped to the table below the operative site** with 3-inch tape.
- **Thigh prepared to expose the anterolateral aspect from greater trochanter to patella.**
- **Landmarks marked:** greater trochanter and lateral femoral condyle — these denote the proximal and distal attachments of the fascia lata (Dwyer and Kreder, 2008).

Graft Harvest for Autologous Pubovaginal Sling

Rectus fascia harvest

- **Pfannenstiel incision 6–7 cm, sited approximately 2 cm above the pubic symphysis** → carried down to the rectus fascia.
- **Mark a 2 × 8 cm graft** on the rectus fascia, transverse or longitudinal.
- **Harvest with scalpel or electrocautery.**
- **If the fascial incision is transverse, stay ≥2 cm away from the pubic symphysis to permit tension-free fascial closure.**
- **Freeing the fascial edges off the underlying rectus muscle aids tension-free closure but may theoretically weaken the fascia** — the trade-off to state.
- **Close the fascia with running No. 1 polydioxanone (PDS). Alternative in patients with minimal subcutaneous fat: figure-of-eight 0-0 polyglactin 910 (Vicryl).**
- **If the fascia cannot be closed adequately — rare — interpose a segment of synthetic mesh or biologic graft.**

Graft preparation

- **Place the graft in 0.9% normal saline, with or without antibiotic.**
- **On a sterile side table, clean off overlying fat and perifascial tissue.**
- **Secure a separate No. 1 PDS or 1-0 polypropylene suture to EACH end of the graft.**
- **Place each suture perpendicular to the sling fibres (transverse harvest only), run it across the end of the graft, and tie down.**
- **Leave sutures long; return the graft to saline until needed.**

Fascia lata harvest

- **Mark a 3 cm longitudinal incision beginning just above the patella, over the iliotibial band.**
- **Dissect to the fascia lata; make two parallel longitudinal incisions 2 cm apart.**
- **Bluntly lift the graft off the underlying muscle; clamp as distally as possible with a right-angle clamp (3–4 cm) and transect, creating one free end.**
- **Secure the free end with No. 1 PDS.**
- **Lift the proximal fascia lata off the muscle belly with a thin malleable retractor, passing it both superficial and deep to the fascia lata to separate it from adipose tissue and muscle fibres.**
- **With the free distal end under tension, use a Crawford fascial stripper to extend the fascial incision proximally and divide it before removal.**
- **Secure a second No. 1 PDS to the other free end; store the sling in saline.**
- **Length: classically a full-length strip of 20 × 2 cm; shorter lengths (8 cm) are now commonly used (Karram and Bhatia, 1990).**

Thigh closure

- **Apply immediate compression to constrict perforating vessels.**
- **Inspect carefully for arterial bleeding before closure.**
- **Irrigate and close in three layers, WITHOUT closing the fascia lata itself.**
- **Apply a compressive wrap and replace the IPC device.**
- **Keep the compressive bandage on for 8 hours postoperatively; encourage early ambulation.**

Pubovaginal Sling Vaginal Dissection

Incision

- Inject 0.9% sterile normal saline into the vaginal epithelium at the mid-urethra and bladder neck → hydrodistention aids tissue dissection.
- Preferred incision: inverted U — it gives excellent exposure of the urethra at the bladder neck and direct access to the endopelvic fascia and thence the retropubic space.
- Top of the incision approximately 2 cm below the urethral meatus; an Allis clamp placed immediately below the meatus improves exposure.
- Arms of the inverted U extend to the level of the bladder neck, located by palpating the Foley balloon.
- 15-blade knife through the vaginal epithelium, staying ABOVE the periurethral and pubocervical fascia — this avoids bleeding and injury to urethra and bladder.
- Alternative: a midline vaginal incision from the bladder base proximally to the mid-urethra distally.
- Create THICK vaginal epithelial flaps with Allis clamp and Metzenbaum scissors; retract with the vaginal ring retractor.
- Keep the arms of the inverted U wide to avoid devascularising the flap.

Entering the retropubic space

- Perforate the endopelvic fascia beneath the easily palpable ischiopubic rami.
- The bladder must be adequately drained before this manoeuvre, and again before passing Stamey needles or larger clamps.
- Metzenbaum scissors angled toward the ipsilateral shoulder, tips pointing upward; perforate directly medial and immediately under the ischiopubic ramus at the superior margin of dissection.
- Perforation is in a SUPEROLATERAL direction; spread the scissors widely to prepare the next step.
- Blunt finger dissection of the retropubic space on either side of the urethra — this connects the infrapubic and retropubic dissection planes.
- Endpoint of adequate dissection: the posterior surface of the pubic symphysis is easily palpable with very little intervening tissue, permitting free movement of the sling and easy tensioning.
- Even with the abdominal fascia closed, simultaneous finger palpation through the abdominal and vaginal incisions should be possible, with gentle medial palpation of the bladder.
- Do NOT attempt aggressive medial mobilisation — this risks bladder injury. Haemostasis with bipolar cautery.

The reoperative case

- After prior urethral suspension or sling, more aggressive SHARP dissection is required.
- In difficult cases the safest plane into the retropubic space is immediately adjacent to the periosteum of the pubis, and dissection should be as sharp as possible to minimise injury to pelvic viscera.

Pubovaginal Sling Placement and Fixation

Needle passage

- Stamey or Cobb-Ragde needles are passed from ABOVE, through the abdominal incision, guided carefully behind the pubis.
- Keep the needles in contact with the pubis until they emerge lateral to the bladder into the vaginal incision.
- Alternative instruments for suture advancement: larger clamps such as tonsil clamps.
- The bladder must be completely drained before passage of any suture passer.

Cystoscopy — mandatory

- **Perform cystoscopy with a 70-degree lens after needle passage**, following the needle course while an assistant moves the needles downward and medially toward the bladder.
- **Cystoscopy is regarded as essential to eliminate the serious complication of sling perforation of the bladder.**
- **If a small bladder injury or transvesical needle passage occurs: remove the needles and repass them MORE Laterally.**
- **Repeat cystoscopy after EVERY subsequent needle passage.**
- **After extravescical passage is confirmed, observe brisk ureteric efflux bilaterally to confirm patency; intravenous methylene blue or sodium fluorescein, or ureteric catheters, may be used.**
- **Replace the Foley catheter for completion of the case.**

Graft fixation and closure

- **Pass the graft suture ends through the Stamey/Cobb-Ragde needle eyelets.**
- **Mark the CENTRE of the graft with a clamp, remove the needles, and bring the suture ends out through the abdominal incision, tagged with hemostat clamps.**
- **Suture the distal aspect of the graft to periurethral tissue with two simple 4-0 polyglactin 910.**
- **Close the vaginal incision watertight with running 2-0 polyglactin 910.**
- **Before final tensioning: close the vagina and REMOVE the weighted speculum — retained, it distorts the field and falsifies the final tension.**
- **Complete any additional procedures, e.g. transvaginal POP repair, BEFORE sling tensioning.**
- **Tie the PDS sutures down ABOVE the rectus fascia, leaving a TWO-FINGERBREADTH gap between the rectus fascia and the knot.**
- **Tension varies with urethral mobility, or is deliberately tighter where permanent retention is intended in a patient who will perform lifelong CIC.**
- **Close Scarpa's fascia with 2-0 polyglactin 910 and skin with subcuticular 4-0 polyglactin 910.**
- **Foley to straight drainage; vaginal packing optional. Conjugated oestrogen cream may be added to the vaginal packing in postmenopausal patients.**

Pubovaginal Sling Postoperative Care

- **Liberal fluids and diet on the night of surgery.**
- **Vaginal packing and urethral catheter both removed on the morning of postoperative day 1.**
- **If unable to void, or PVR is elevated: discharge with an indwelling Foley and return within 5 days for a repeat trial of voiding.**
- **Avoid lifting more than 10 pounds and abstain from intercourse for at least 6 weeks.**
- **Intercourse resumes only after the surgeon confirms vaginal healing on physical examination.**

KEY POINTS: PUBOVAGINAL SLING OPERATIVE PROCEDURE

- **Tension-free rectus fascial closure requires maintaining ≥ 2 cm from the pubic symphysis. If undermining the fascial edges does not mobilise enough fascia, interpose synthetic mesh or a biologic graft to bridge the gap.**
- **Ensure the bladder is empty before perforating the endopelvic fascia, before retropubic dissection, and before passing needles.**
- **Cystoscopy is imperative after the initial trocar passage AND after every subsequent pass, to confirm integrity of bladder and urethra.**

- **Sling tensioning should be intentionally LOOSE to minimise obstruction.**

OUTCOMES OF AUTOLOGOUS PUBOVAGINAL SLINGS FOR PREDOMINANTLY STRESS URINARY INCONTINENCE

Autologous Pubovaginal Slings

Primary SUI and the complex population

- **Reintroduced in the late 1970s (McGuire and Lytton, 1978)**, the autologous PVS has remained **reproducible and effective at long-term follow-up**.
- The **earliest series** (McGuire et al., 1987; Blaivas and Jacobs, 1991) deliberately included a **complex population** — **pelvic radiation, diabetes, spinal cord injury, pelvic trauma**.
- **Even patients with "proximal urethral function loss", defined as MUCP <4 cm H₂O, achieved a high cure rate** — the striking demonstration that **severe ISD is not a contraindication to PVS**.
- **Concomitant anterior colporrhaphy with PVS gives excellent results in grade 3–4 cystocele with SUI** (Cross et al., 1997).

Recurrent SUI

- **Groutz et al. (2001)**, comparing primary with recurrent SUI (the latter after one to three failed anti-incontinence procedures): **cure was significantly higher in primary SUI, but ALL patients in both groups were cured or improved**.
- **Richter et al. (2001)**, long-term QoL after PVS, most having recurrent SUI after failed bladder neck suspension: **the large majority felt the sling improved QoL and would undergo surgery again**; despite postoperative voiding dysfunction, **all were satisfied with SUI resolution and none elected urethrolisis**.
- **Petrou and Frank (2001)**: **high cured or improved rate for REPEAT PVS after failed suburethral sling**.
- **Athanasopoulos et al. (2011)** — one of the earliest and largest series of **PVS after failed MUS**; roughly a third had prior MUS, **all underwent partial sling excision at the time of PVS**, with a **high cured or markedly improved rate**.

Landmark Trial — SISTER (Albo et al., 2007)

- **Stress Incontinence Surgical Treatment Efficacy Trial: multicentre randomised trial of rectus fascia PVS (326) versus Burch colposuspension (329)**.
- **Two primary outcomes: resolution of overall UI** — no subjective or objective incontinence, **increase of <15 g on 24-hour pad weight**, and no medical or surgical treatment for incontinence; and **resolution of SUI** — no subjective SUI and a **negative cough stress test**.
- **Result: PVS gave a higher SUI cure rate (47% vs 38%, $p < 0.001$), but significantly higher rates of postoperative voiding dysfunction and cystitis**.
- **All 20 subsequent procedures for bladder outlet obstruction (BOO) occurred in the PVS group**.
- **Long-term extension (Brubaker et al., 2012): 5-year continence 31% (PVS) versus 24% (Burch), $p = 0.002$. Satisfaction at 5 years higher after PVS (83% vs 73%), declining with time in both arms but remaining higher in the sling group. Adverse event rates were similar (Burch 10% vs PVS 9%)**.
- **Interpretation: the low absolute success rates reflect a STRICT definition of success and vigilant follow-up, not a poor operation**.
- **The key conceptual observation: subjective satisfaction diverges from strict objective cure** — **patients can be satisfied without being completely dry**.

Technical modifications

- **Mason and Roach (1996):** a shorter 4 × 2 cm modified rectus fascia PVS used successfully in ISD.
- **Ghoniem (1991):** a modified PVS fixing a 7 × 2.5 cm graft to the pubocervical and periurethral ligaments in a FOUR-QUADRANT manner; with longer follow-up (Hassouna and Ghoniem, 1999), cure was high and every documented failure had postoperative UUI or severe storage symptoms — reinforcing that failure of PVS is usually a storage-symptom problem, not a stress problem.

Fascia lata outcomes

- **Intermediate-term outcomes appear comparable** to rectus fascia in retrospective series.
- **Beck et al. (1988):** high cure by postoperative CST in recurrent SUI. **Govier et al. (1997) and Brown and Govier (2000):** lower, questionnaire-derived satisfaction-based cure — illustrating that the measure chosen dictates the reported success rate.
- **Harvest-site morbidity is the differentiator:** all patients had thigh pain at one to two weeks in one series, with a minority having persistent pain at six weeks. **Latini et al. (2004)** at one week: localised harvest-site numbness in a fifth, pain in a small minority, and tendinitis in the harvest leg in a few — yet most reported a positive effect on life. This remains the ONLY fascia lata PVS study with mean follow-up ≥48 months.
- **Peng et al. (2020),** rectus fascia versus fascia lata: similar freedom from SUI at 1 month and 1 year; wound complications made up almost a third of postoperative complications in the rectus group; the only significant difference was lower estimated blood loss favouring fascia lata.

Cochrane systematic review (Saraswat et al., 2020)

- 34 trials, most with follow-up <24 months.
- PVS versus MUS (14 trials): PVS are probably NO BETTER, and may be LESS effective, than MUS for continence at 1 to 5 years. Not all PVS were autologous.
- MUS was associated with shorter operating times, fewer perioperative complications EXCEPT bladder perforation, and less postoperative voiding and storage dysfunction.
- PVS versus PVS (9 trials): results could NOT be pooled owing to clinical heterogeneity of sling types and materials.

PVS with mesh complication surgery

- **Shah et al. (2013):** PVS at the time of urethral repair for MUS exposure — continence in most, no major intraoperative complications, and complete resolution of mesh-related symptoms in all.
- **Milose et al. (2015):** staged autologous sling after prior urethral diverticulectomy, urethral mesh exposure and/or urethrovaginal fistula — cure of SUI in about two-thirds of the whole cohort, but NONE of the previously diverticulectomised or mesh-excised subgroup was completely continent. The damaged urethra is the poor-prognosis group.

Summary of outcomes

- Success rates after PVS are typically high but range very widely (24–97%), depending on the patient population and the definition of success.
- There is NO direct correlation between accepted objective and subjective measures of cure for anti-incontinence procedures (Padmanabhan and Nitti, 2006).
- The commonest cited reason for failure is other urinary storage symptoms; UUI at follow-up is a common cause of dissatisfaction.
- Postoperative de novo UUI: 2–22%.
- **Yang et al. (2021):** prior POP surgery predicted de novo urgency after autologous PVS; the overall incidence of treatment for de novo urgency or UUI was 21%, and age, body mass index (BMI), pelvic

irradiation, anticholinergic use, "over-tight" technique and concomitant surgery were NOT associated with it in that cohort.

Allograft, Synthetic and Xenograft Pubovaginal Slings

- Outcomes of allograft PVS are summarised in Table 118.2, of xenograft PVS in Table 118.3; detailed discussion of these and of randomised trials of synthetic bladder-neck PVS is in the online version of the chapter.

TABLE 118.1 LONG-TERM (≥48 MONTHS) OUTCOMES OF AUTOLOGOUS PUBOVAGINAL SLINGS

- **Columns:** lead author and year · number · type (rectus fascia or fascia lata) · mean or median age · genuine stress urinary incontinence (GSI) proportion · mean or median follow-up in months · cured or improved · method of outcome assessment · de novo urgency or UUI (%) · persistent urinary retention (%).
 - **What it demonstrates:** across series with follow-up from roughly four years to well over a decade, **autologous PVS sustains cure or improvement in the majority**, but **reported success falls as follow-up lengthens and as the assessment method becomes more objective** — chart review and subjective report give the highest figures, validated questionnaire with cough stress test the lowest.
 - **De novo urgency or UUI and persistent retention are both recorded but inconsistently reported**, several entries being not recorded.
 - **Both rectus fascia and fascia lata appear in this long-term table**, several series using either.
 - **Annotations:** one series included patients with nocturia and positional voiding in its de novo urgency figure; one series comprises patients who had **failed a prior MUS, with concomitant or previous MUS or mesh removal** — a distinct and harder population.
 - **Key:** CR, chart review · CST, cough stress test · FL, fascia lata · GSI, genuine stress urinary incontinence · NR, not recorded · R, rectus fascia · UUI, urge urinary incontinence · VD, voiding diary · VQ, validated questionnaire.
-

TABLE 118.2 OUTCOMES OF ALLOGRAFT PUBOVAGINAL SLINGS

- **Columns:** lead author and year · number · tissue · age · GSI (%) · mean or median follow-up in months · cured or improved · assessment of outcome · de novo urgency or UUI (%) · persistent urinary retention (%).
- **Tissue types are the examinable content: fresh frozen (FF), freeze dried (FD), solvent dehydrated (SD), and dermal graft (DG)** — the four processing methods, which is exactly the point that **processing technique varies and may disrupt microstructure and graft strength**.
- **What it demonstrates:** results with allograft are **markedly heterogeneous and generally inferior and less durable than autologous fascia**, with **follow-up mostly short (commonly one to three years)** and cure rates spanning a very wide range at similar follow-up. **De novo urgency is variably reported and can be substantial; persistent retention is generally low**.
- One series stratified outcomes by age above and below 80 years.
- **Footnote: retention is defined as catheterisation for longer than 2 weeks** — the operative definition to hold.
- **Key:** CR, chart review · CST, cough stress test · DG, dermal graft · FD, freeze dried · FF, fresh frozen · GSI, genuine stress urinary incontinence · NR, not

recorded · SD, solvent dehydrated · UDI, Urogenital Distress Inventory · UUI, urge urinary incontinence · VD, voiding diary · VQ, validated questionnaire.

TABLE 118.3 OUTCOMES WITH XENOGRAFT PUBOVAGINAL SLINGS

- **Columns:** lead author and year · number · type · age · follow-up in months · cured or improved · assessment of outcome · de novo urgency (%) · persistent urinary retention (%).
 - **Xenograft types named:** Pelvicol (porcine dermis), small intestinal submucosa (SIS), and Xenform Matrix.
 - **Footnote:** Xenform Matrix is acellular BOVINE dermis — the one xenograft in the table that is not porcine.
 - **What it demonstrates:** early short-term results with xenograft appear good, but **cure falls substantially with longer follow-up** — the same material reported at three years and at ten years shows a **steep decline in dryness**, which is the argument against xenograft durability. De novo urgency and persistent retention are low but sparsely reported.
 - **Key:** CR, chart review · CST, cough stress test · NR, not recorded · SIS, small intestinal submucosa · SUI, stress urinary incontinence · VQ, validated questionnaire.
-

OUTCOMES OF AUTOLOGOUS PUBOVAGINAL SLING IN PATIENTS WITH MIXED URINARY INCONTINENCE

- **SUI with concomitant urgency or UUI is a complicated treatment problem**, usually requiring **combined pharmacological therapy and surgery**.
- **Medical therapy for MUI resolves the urgency component in only about two-thirds of patients** (Nordling et al., 1979; Stephenson and Mundy, 1994).
- **Pharmacological therapy does not address the bladder outlet**, and therefore **cannot be expected to achieve dryness where there is a significant stress component**.
- **Anti-incontinence surgery may cure, improve, OR aggravate storage symptoms** — the three-way outcome is the point.
- **Even an uncomplicated sling may produce de novo urgency or UUI**. This aspect of anti-incontinence surgery is fundamentally unpredictable and is a major reason for patient dissatisfaction.
- **No consistent clinical or videourodynamic parameter predicts urgency outcomes after PVS in MUI**.
- **Residual urgency after PVS behaves similarly to de novo urgency**, and increasing episodes of urgency and UUI may correlate with surgical failure.
- **Overall the PVS remains effective in MUI**, with cure rates similar to those in pure SUI.

OUTCOMES OF AUTOLOGOUS PUBOVAGINAL SLING FOR URETHRAL RECONSTRUCTION

- Autologous PVS has an important role in reconstruction after **anatomical damage with tissue loss**, across a spectrum from the **mild — urethrovaginal fistula and urethral diverticulum** — to the **severe — traumatic absence of the urethra or bladder neck**.
- Full discussion is in the online version of the chapter.

KEY POINTS: PUBOVAGINAL SLING OUTCOMES

- Cure rates after autologous PVS vary widely with the definition of success and the outcome measured; de novo urgency and UI rates are likewise variable.
- No preoperative risk factors consistently predict outcomes after PVS.
- Cure of RECURRENT SUI by PVS is excellent. In randomised trials, porcine dermis gave significantly inferior long-term cure than autologous PVS.
- Treatment of MUI is complicated and may require combined anticholinergic therapy and surgery.
- PVS is effective for stress-induced DO, with cure similar to PVS for SUI. However, preoperative DO may be associated with reduced postoperative QoL and lower rates of urgency resolution.
- Autologous PVS is a valuable adjunct providing both continence and robust tissue coverage in urethral reconstruction — for urethral fistula, diverticulum, complications after previous MUS, or the devastated urethra.

COMPLICATIONS OF PUBOVAGINAL SLINGS**Voiding Dysfunction From Bladder Outlet Obstruction After Pubovaginal Sling***Incidence*

- **Voiding dysfunction after PVS: 2.5–35%**, exceeding rates after Burch colposuspension.
- **SISTER confirmed significantly more postoperative voiding dysfunction, UTI and postoperative UII after PVS than after Burch.**
- **First AUA SUI Clinical Guidelines Panel meta-analysis (Leach et al., 1997):** urinary retention beyond 4 weeks after PVS occurs in 8%, and the risk of PERMANENT retention generally does not exceed 5%.
- **Morgan et al. (2000):** prolonged retention in only 2.4% at four-year follow-up — taken as the appropriate rate when slings are not intentionally overtensioned.

Presentation — deliberately variable

- **Purely voiding symptoms:** decreased force of stream, positional emptying, elevated PVR, urinary retention.
- **Purely storage symptoms:** urgency, UII. Or a combination of both.
- **Obstruction may also present as recurrent UTI, prolonged suprapubic pain, or painful voiding EVEN WHERE emptying is complete and PVR is low** — the trap in relying on PVR alone.
- **Nitti et al. (2002):** a sixth of patients requiring urethrolisis after PVS had NO voiding symptoms and no retention.
- **Cross et al. (1998b):** persistent postoperative urgency or UII (around a quarter of patients) is a COMMONER presentation of obstruction than frank retention (around 8%).

Mechanism — the tension dilemma

- **Iatrogenic obstruction is usually technical:** sutures or sling material placed under excessive tension.
- **There is a fine line between inadequate and excessive tension, and no consistent way of determining ideal tension for the individual patient.**
- **Too loose → inadequate support of bladder neck and proximal urethra → persistent SUI.**
- **Too tight → excessive elevation of the bladder neck toward the pubic bone → hypersuspension, overcorrection of the urethrovesical angle → obstruction.**

Host factors predisposing to postoperative emptying failure

- **Pre-existing voiding dysfunction.**

- **Subclinical impaired detrusor contractility** — becomes symptomatic as **relative obstruction once outlet resistance rises** (Weinberger and Ostergard, 1996).
- **Dysfunctional voiding** — **failure of relaxation of the external (striated) urethral sphincter** (FitzGerald and Brubaker, 2001).
- **Habitual voiding by abdominal straining** — theoretically problematic because **the sling provides DYNAMIC resistance that increases with intra-abdominal pressure**, so the straining meant to empty the bladder instead tightens the outlet.
- **A thorough history establishing chronology, character and bother of the predominant symptom is invaluable**, given the variability of presentation.

Evaluation of suspected post-sling obstruction

- **Physical examination** may show **abnormal urethral angulation**, a **foreshortened non-pliable vagina**, or a **non-mobile urethra** — but **hypersuspension may not be evident on examination**.
- **PVR is very important**, though **NO clear cut-off value for obstruction exists** (Siddighi and Karram, 2007).
- **Cystoscopy** to exclude **bladder pathology**, **sling perforation**, and a **hypersuspended urethra**.
- **Videourodynamics in selected cases at the physician's discretion** — but the **classic high-pressure, low-flow pattern pathognomonic of obstruction is frequently ABSENT in females**.
- **The single most important criterion for deciding to incise a sling or perform urethrolisis is the TEMPORAL RELATIONSHIP between onset of symptoms and the operation** — not a urodynamic number.

Attempts to predict postoperative voiding dysfunction

- **Weinberger and Ostergard (1996)**: impaired detrusor contractility predicted retention after synthetic PVS.
- **Miller et al. (2003)**: females voiding with minimal or no rise in detrusor pressure had significantly higher retention risk after allograft PVS; **none with a normal detrusor contraction developed retention**.
- **McLennan et al. (1998)**: did not confirm the association.
- **Mitsui et al. (2007)**: a **trend toward prolonged CIC** in those with **preoperative PVR >100 mL** or **maximum flow rate (Qmax) ≤20 mL/s** — a substantial minority needed CIC for up to 40 months.
- **Lemack et al. (2008)**, within **SISTER**: urodynamic findings did **NOT** predict postoperative voiding dysfunction.
- **Practice point**: low detrusor pressure and Valsalva voiding preoperatively should **NOT** exclude a patient from an anti-incontinence procedure.

Prolapse as a cause of post-sling obstruction

- **POP uncorrected at surgery**, or **arising de novo afterwards**, is a **key factor**.
- **Prolapse of sufficient size may kink, angulate or externally compress the urethra**, and **POP in ALL THREE compartments (apical, anterior, posterior) must be excluded as a cause of obstruction**.
- **Kobashi et al. (2002)**, combined cystocele repair and PVS using a **single piece of cadaveric fascia**: at one year, **only one patient developed persistent obstruction requiring urethrolisis**.
- **Barnes et al. (2002)**, concurrent PVS with prolapse repair in **grade 3 or 4 POP with occult SUI**: **no permanent retention**, two **de novo UUI**, and **preoperative UUI resolved in nearly half**. Conclusion: **concurrent surgery had little negative effect on postoperative emptying**.
- **There is a paucity of literature on the emptying and voiding effects of concurrent PVS and prolapse repair**.

The causality problem

- Preoperative storage symptoms after PVS may **resolve, improve, remain unchanged, or worsen** — and may **appear de novo**.

- **Postoperative urgency and UI are strongly related to failure, yet no preoperative risk factor consistently predicts them.**
- **Consequently assigning causality — and deciding whether to investigate and treat BOO rather than the storage symptoms themselves — is genuinely difficult** (Gomelsky et al., 2022). Two reasons:
- **The effect of sling loosening or urethrolisis is unpredictable: storage symptoms may improve or persist, and stress continence may be lost in a patient who is currently dry.**
- **In a patient who does not undergo loosening for incomplete BOO, the long-term consequence for the detrusor is unknown** — the effect of long-standing BOO on the bladder is well characterised in males but not in females.
- **Therefore a detailed risk-benefit discussion about sling loosening or urethrolisis must be held with the patient.**

Surgical Management of Voiding Dysfunction After Pubovaginal Sling

Timing — the governing rule

- **Transient retention is common; most females return to spontaneous efficient voiding within the first 10 days.**
- **Obstruction after autologous PVS usually improves or resolves with time, so most clinicians wait up to 3 MONTHS before considering surgical intervention.**
- **Initial conservative management: temporary indwelling catheter · CIC · timed and double voiding · biofeedback · PFMT · anticholinergic therapy.**

Early intervention in complete retention

- **If in complete retention, revision may be undertaken sooner.**
- **Within the first 6 weeks after autologous PVS the sling may be loosened in theatre under spinal or general anaesthesia: insert a cystoscope into the bladder, then apply gentle inferior pressure to the urethra.**
- **This manoeuvre is NOT advised with synthetic slings.**
- **Urethral dilation with downward traction after autologous PVS has only anecdotal support, and results range from temporary relief to worsening urethral rigidity from periurethral fibrosis.**
- **Transurethral resection or incision of the bladder neck is likely to FAIL, because the sling is EXTRALUMINAL; it also risks damage to sphincter or bladder neck and periurethral fibrosis, causing worsened incontinence or bladder neck contracture** (Ghoniem and Elgamasy, 1995).
- **After 6 weeks, or when conservative measures fail: sling incision or formal urethrolisis is indicated.**

Urethrolisis — approaches and results

- **Three approaches: retropubic, transvaginal, suprameatal. Overall success 65–93%.**
- **Foster and McGuire (1993): transvaginal urethrolisis succeeded in only half of PVS obstructions — lower than for needle suspension or retropubic urethropexy. Their explanation: transvaginal LATERAL dissection is insufficient to relieve the DIRECT SUBURETHRAL COMPRESSIVE force of a sling.**
- **Petrou et al. (1999): the suprameatal approach is superior to transvaginal because it allows access to and division of the LATERAL WINGS of the sling; most of their patients succeeded.**
- **Petrou and Young (2002): retropubic urethrolisis resolved obstruction in most, though a minority subsequently developed SUI. Carr and Webster (1997): significant or complete resolution in the large majority after retropubic urethrolisis.**
- **Recurrent SUI after formal urethrolisis: 0–19%. Outcomes of urethrolisis and sling revision for BOO are summarised in Table 118.4.**

Sling incision versus formal urethrolisis

- **Sling incision gives comparable success (84–100%) with SHORTER operative time and LESS morbidity than formal urethrolisis** — the practical argument for trying it first.
- **Midline or lateral sling incision WITHOUT graft interposition has been reported successful by several authors.**
- **Sling incision WITH autologous graft interposition between the cut ends** was described to prevent recurrent SUI (Ghoniem and Elgamasy, 1995), **but others report suboptimal outcomes.**
- **Vaginal wall interposition relieved obstruction in almost all patients in two series — yet SUI recurred in a quarter of each.** The interposition does not reliably protect continence.

The four takeaways from the sling-incision literature

- **A LATERAL incision theoretically minimises urethral injury where the sling can be identified — but the dissection plane between urethra and sling cannot be established consistently or safely.**
- **Approximately 20% develop recurrent SUI, though not all seek further treatment.**
- **A minority still benefit from subsequent formal urethrolisis after an initial failure to relieve BOO.**
- **No preoperative or urodynamic parameter consistently predicts success or failure of urethrolisis.**

Contested predictors of urethrolisis failure

- **Foster and McGuire (1993): DO predicted higher failure — contradicted by later studies.**
- **Carr and Webster (1997): the only predictor of success was NO PRIOR urethrolisis.**
- **Nitti and Raz (1994): failure related to higher PVR.** None of these has been confirmed by others.

Causes of persistent symptoms and repeat urethrolisis

- **Four causes of persistent storage or voiding symptoms after urethrolisis: persistent or recurrent obstruction · detrusor overactivity · impaired detrusor contractility · learned voiding dysfunction.**
- **Recurrent obstruction may arise from periurethral fibrosis and scarring, or from intrinsic urethral damage caused by previous urethrolyses.**
- **The commonest reason for failure is likely INSUFFICIENT dissection and lysis of the urethra — an inadequate operation, not an unfavourable patient.**
- **Scarpero et al. (2003): repeat urethrolisis after initial failure resolved obstruction in the large majority (92%), using transvaginal or retropubic approaches according to the clinical situation. But storage symptoms resolved in only a small minority (12%), and most (69%) continued to need anticholinergic therapy; SUI recurred in 18%.**
- **Repeat urethrolisis is supported after initial failure, or where the aggressiveness of the first dissection is in doubt. Retropubic urethrolisis may be considered after a failed aggressive TRANSVAGINAL urethrolisis — the escalation pathway.**

Refractory storage symptoms after urethrolisis

- **Refractory storage symptoms affect more than half of patients after urethrolisis, with real impact on satisfaction and quality of life (Starkman et al., 2008a).**
- **There are no predictors of this outcome, though preoperative DO may suggest an increased likelihood of refractory OAB. There are no established guidelines for optimal management.**
- **Starkman et al. (2007), in females with UII after pelvic surgery (mostly PVS): the large majority reported greater than 50% symptom improvement after sacral neuromodulation (SNM), and some became continent. No significant difference in response by age, symptom duration, type of surgery, or urodynamic parameters — i.e. no way to select responders in advance.**
- **Starkman et al. (2008b), in patients who had undergone at least one urethrolisis: most had a favourable response at test stimulation, and ALL of those responders remained significantly improved after**

implantation of the implantable pulse generator — test stimulation reliably predicted the durable outcome.

- **Alongside anticholinergics, SNM should be considered for de novo or refractory urgency and UI after urethrolisis.**
- **OnabotulinumtoxinA** has also been used for persistent and de novo OAB symptoms after sling surgery. Miotla et al. (2016), comparing idiopathic OAB with a MUS against de novo OAB after MUS, both given 100 U: at 12 weeks roughly 40% of each group were completely dry, with no significant difference between groups; both had significant improvement in daily voids and incontinence episodes, and mean voided volume increased by more than 90 mL. **Critical caveat: these patients did NOT undergo sling revision first.**

TABLE 118.4 RESULT OF THERAPY FOR BLADDER OUTLET OBSTRUCTION AFTER PUBOVAGINAL SLING SURGERY

- **Columns:** lead author and year · number · management · mean time to treatment (months) · success rate (%) · recurrent SUI (%) · subsequent treatment.
- **Two management categories are compared: formal urethrolisis (transvaginal, retropubic, infrapubic or suprameatal) and sling lysis or incision.**
- **Mean time to treatment varies enormously across series — from under two months to over two years — reflecting the absence of any agreed trigger for intervention.**
- **Within a single series comparing approaches, retropubic urethrolisis outperformed vaginal, which outperformed infrapubic** (Carr and Webster, 1997).
- **Transvaginal urethrolisis for post-PVS obstruction performs least well**, consistent with the mechanistic argument that lateral dissection cannot relieve direct suburethral compression.
- **Sling lysis series report consistently high success**, supporting its use as the lower-morbidity first option; **recurrent SUI is the price**, reported from none to around a third.
- **Subsequent treatments recorded are periurethral bulking, repeat PVS, and MUS — the salvage options for post-lysis recurrent SUI.**

KEY POINTS: VOIDING DYSFUNCTION AFTER PUBOVAGINAL SLING

- **Obstruction, DO and impaired detrusor contractility are all manifestations of voiding dysfunction from iatrogenic outlet obstruction after PVS.**
- **Permanent urinary retention is usually $\leq 5\%$.**
- **BOO after PVS presents more frequently with persistent urgency and UI (8–25%) than with frank retention.**
- **The temporal relationship between the anti-incontinence surgery and the onset of voiding dysfunction is the key to diagnosis;** urodynamics may aid diagnosis and treatment planning.
- **Transient retention usually resolves within 10 days, and obstruction from a PVS may improve or resolve with time — therefore most surgical intervention should be postponed until at least 3 months after surgery.**
- **Surgical management of post-PVS BOO succeeds in 65–93%; recurrent SUI after urethrolisis is 0–19%.**
- **No preoperative variable predicts success or failure of urethrolisis. OAB symptoms are refractory in around half of affected patients after urethrolisis and account for a significant proportion of reported failures.**

Pubovaginal Sling Perforation and Exposure

- **Incidence depends on sling composition. Synthetic slings perforate into the urethra 15 times more often, and are exposed in the vagina 14 times more often, than autologous, allograft and xenograft slings** (Blaivas and Sandhu, 2004), based on the **meta-analysis of 287 peer-reviewed articles** (Leach et al., 1997).
- **Subsequent studies confirm that most perforations and exposures involve synthetic slings, particularly WOVEN POLYESTER slings.** Cases of autologous and allograft perforation and exposure have been published, but they are rare.
- **Most urethral perforations are diagnosed between 1 and 18 months after the original surgery, with a mean of 9 months.** Perforation from synthetic slings is discussed in Chapter 119.
- **Urinary tract perforation by an autologous PVS is rare: only four documented cases of bladder and urethral erosion in the peer-reviewed literature.**
- **Contributing factors:** missed bladder perforation after Stamey needle passage • excessive tension • traumatic urethral instrumentation after sling placement — e.g. for clot evacuation or surveillance cystoscopy.
- **Management:** urethral perforation → excise the perforated portion of the sling and repair the urethra. Bladder erosion → treat endoscopically. Rarely, a Martius flap is required for complex cases.

Non-urollogic Pubovaginal Sling Complications

- **Most PVS complications are urinary tract related, but significant non-urollogic complications occur.**
- **2009 update of the AUA guideline for SUI surgery (Appell et al., 2009), by meta-analysis: the commonest non-urollogic complications are, in order, PULMONARY, CARDIOVASCULAR, NEUROLOGIC and GASTROINTESTINAL (bowel injury).**
- **Across all anti-incontinence procedures — retropubic and transvaginal suspensions, anterior colporrhaphy and PVS — the estimated death rate is approximately 3 per 10,000 procedures.**
- **Anger et al. (2007), sling surgery (mostly MUS) in Medicare beneficiaries aged 65 and over: within 3 months, about an eighth developed surgical or urollogic complications, and a third were diagnosed with UTI.** At one year, non-urollogic complications recorded were **pulmonary (the commonest), cardiac, bowel injury or obstruction, and thromboembolic.** On multivariate analysis, non-White race was a risk factor for both urollogic and non-urollogic complications, and women aged 65–69 were significantly **LESS** likely to have non-urollogic complications, or to need treatment for BOO or recurrent incontinence, than those over 75.
- **Chughtai et al. (2016), in over 21,000 female Medicare beneficiaries: adverse events within 90 days were rare, the sole exception being urinary retention, recorded in about one in nine.**
- **SISTER complication data: serious adverse events 13% after autologous PVS versus 10% after Burch.** In the sling group, deep venous thrombosis (DVT) and serious bleeding each occurred in well under 1%, and serious wound complications in a small minority.
- **Wound complications specific to autologous harvest: the commonest are harvest site infection, seroma, and incisional hernia.** Blaivas and Chaikin (2011), in over 500 patients: wound infection and incisional hernia each around 1%; Morgan et al. (2000) reported a similarly sub-1% hernia rate.
- **Incisional hernia from rectus fascia harvest, though rare, is a serious complication that usually requires surgical correction — the price of the gold-standard material.**

KEY POINTS: COMPLICATIONS OF PUBOVAGINAL SLINGS

- Perforation incidence depends on sling material: synthetic slings are 15 times more likely to perforate into the urethra and 14 times more likely to be exposed in the vagina than other materials. **FOR THIS REASON, SYNTHETIC PVS IS NO LONGER USED FOR SUPPORT AT THE BLADDER NECK.**
- Vaginal exposure after a synthetic sling typically presents with vaginal discharge, pain, suprapubic pain and recurrent UTI, and may require subtotal or total sling removal.
- The incidence of recurrent SUI after revision of a synthetic PVS for urethral perforation is 44–100%, and treatment often involves a second, **AUTOLOGOUS, PVS.**
- Perforation or exposure after autologous PVS is rare.

MID-URETHRAL SLINGS**Mechanics of the Mid-Urethral Sling***Theoretical basis*

- **Integral theory (Petros and Ulmsten, 1990):** continence depends on adequate function of the pubourethral ligaments, the suburethral vaginal hammock, and the pubococcygeus muscle. Injury to any one — from surgery, parturition, ageing or hormonal deprivation — impairs mid-urethral function and causes SUI.
- **Conceptually critical:** urethral hypermobility is a **SIGN** of damage to the normal supporting structures — **NOT** a cause of SUI.

Imaging observations

- Preoperative ultrasonography and MRI show the proximal urethra in females with SUI is often **OPEN AT REST.**
- Regardless of urethral mobility, movement of the anterior and posterior urethral walls during stress produces a **SHEARING** effect that opens the urethra and causes leakage.

How the MUS works — three proposed mechanisms

- It impedes movement of the posterior urethral wall above the sling, redirecting its motion anteroinferiorly or anteriorly.
- Inward movement of the posterior urethral wall against the sling compresses and narrows the urethral lumen.
- Some authors propose the MUS additionally **KINKS** the urethra dynamically during stress events, as seen on ultrasound (Lo et al., 2001).
- The unifying concept: securing the posterior urethral wall, with or without compression during stress.

Anchoring

- Retropubic MUS anchors to the **ENDOPELVIC FASCIA**; transobturator MUS anchors to the **OBTURATOR INTERNUS AND EXTERNUS** muscle and fascia.
- Ideally either is placed **LOOSELY** at the mid-urethra; function does not require the sling to be tight.
- Over time the mesh becomes fixed at its anchoring site and provides support along its **ENTIRE** course inferior to the pubic symphysis and ischiopubic rami — not merely in the midline behind the urethra.
- This broad support is the likely explanation for why 50–93% remain continent after MUS **INCISION** for obstruction — cutting the midline does not abolish the whole support.

Urethral Mobility as a Prognostic Factor

Retropubic and transobturator MUS — the standard rule

- **Lack of urethral mobility BEFORE surgery predicts lower SUI cure rates.**
- **Decreased urethral mobility AFTER surgery is associated with persistent incontinence.**
- **Conversely, persistent significant urethral mobility postoperatively is NOT associated with a worse outcome.**
- **Rationale: absent mobility signifies a FIXED urethra and potentially ISD, and females without hypermobility do not respond as well to MUS surgery.**
- **The ideal combination is a loosely placed MUS with a MOBILE urethra — allowing compression during Valsalva while remaining non-obstructive at rest.**

SIMS — the exception

- **Unlike retropubic and transobturator MUS, RESTRICTION of urethral mobility after SIMS may be associated with BETTER outcomes.**
- **Martan et al. (2009), using perineal ultrasonography, cough test and validated questionnaires: the sling restricted urethral mobility, and a greater degree of restriction predicted a higher likelihood of cure — but the restrictive effect weakened within the first 3 months.**
- **Spelzini et al. (2013): outcomes after SIMS were WORSE in those with persistent urethral hypermobility.**
- **Proposed explanation: SIMS efficacy depends on restricting mobility, and that restriction is required to secure the posterior urethral wall, achieve adequate kinking, or compress the lumen.**

Does the Mid-Urethral Sling Obstruct? — The Controversy

- **Despite tension-free placement, it is disputed whether tension or resistance is ultimately created at the bladder outlet.**
- **Evidence AGAINST obstruction: Meschia et al. (2001), over 400 retropubic MUS: increased voiding time postoperatively, but no difference in flow rate, urethral closure pressure (UCP) or functional urethral length. Lo et al. (2001): no statistical difference in average flow rate (Qave), Qmax, PVR, MUCP or functional urethral length.**
- **Evidence FOR obstruction: Gateau et al. (2003): consistent decrease in Qmax, increased mean detrusor pressure at maximum flow (PdetQmax), increased mean urethral resistance and elevated PVR — concluding retropubic MUS produces obstructive changes at the bladder outlet. Sander et al. (2002), one year after retropubic MUS: both subjective and objective voiding-phase changes — most patients reported more difficult voiding, with significant falls in Qmax, corrected Qmax and Qave, and significantly increased PVR, though never exceeding a quarter of capacity.**

Anatomy of the Retropubic Mid-Urethral Sling

- **Anatomical landmarks for positioning retropubic and transobturator MUS are covered in the online version of the chapter.**

Mid-Urethral Sling Materials

Historical sequence

- **Ulmsten and Petros (1995) initially used several materials: Mersilene, Gore-Tex, Teflon, and Lyodura (cadaveric dura mater).**
- **Because of complications with other synthetic materials in the general surgical HERNIA literature, Ulmsten et al. (1998) settled on a polypropylene multifilamentous woven mesh.**

The modern standard

- **Almost all current MUS products are soft, loosely woven, POLYPROPYLENE MONOFILAMENT mesh with pore size exceeding 75 µm.**
- **Why >75 µm matters: it permits migration of host inflammatory cells (leukocytes and macrophages) into the mesh for infectious surveillance and host wound healing (imbibition and inosculation).**
- **The same material is optimal for inciting fibrous tissue ingrowth.**
- **This is TYPE 1 MESH in the Amid classification, described in the general surgical literature as favourable for its mechanical properties — stretch and elasticity.** Further discussion is in Chapter 119.

Transobturator product evolution — a cautionary sequence

- **UraTape (Mentor-Porgés) — the FIRST transobturator MUS (Delorme et al., 2003): polypropylene, MICROPOROUS, with a central SILICONE core.**
- **Replaced by ObTape because of a high rate of vaginal exposure, likely due to the silicone core — but vaginal exposures also occurred with ObTape, probably because of its semi-microporous nature (<50 µm).**
- **Second-generation Aris transobturator tape: larger 200 µm pore size, allowing improved tissue ingrowth with LESS encapsulation.**
- **BioArc — unique in having a BIOLOGIC (porcine dermis, InteXen) graft sutured at either end to polypropylene mesh.**
- **Many retropubic and transobturator MUS are off the market; all available products are monofilament, MACROPOROUS, woven polypropylene.**

KEY POINTS: MECHANICS, ANATOMY, AND MATERIALS OF MID-URETHRAL SLINGS

- **The integral theory holds that continence depends on adequate function of the pubourethral ligaments, suburethral vaginal hammock and pubococcygeus muscle; injury to any of the three from surgery, parturition, ageing or hormonal deprivation impairs mid-urethral function and causes UI.**
- **Urethral hypermobility is a sign of damage to the normal urethral supporting structures, NOT a cause of SUI.**
- **Initial MUS products used materials with smaller pore sizes; the majority of current MUS are loosely woven, MACROPOROUS, MONOFILAMENT polypropylene.**

MID-URETHRAL SLING OPERATIVE PROCEDURES**Mid-Urethral Sling Patient Counselling**

- **Because MUS surgery implants a synthetic prosthesis, counsel thoroughly about the PERMANENT nature of these products and about unique and sometimes serious complications, though infrequent.**
- **AUA/SUFU Clinical Principle: before selecting a synthetic MUS, physicians MUST discuss the specific risks and benefits of mesh AS WELL AS the alternatives to a mesh sling (Kobashi et al., 2017, 2023).**
- **As with PVS, counsel about transient and permanent voiding dysfunction: postoperative difficulty emptying the bladder, and de novo or worsened urinary storage lower urinary tract symptoms (LUTS).**

Mid-Urethral Sling Anaesthesia, Patient Positioning and Preparation

- **MUS may be performed under local (with or without sedation), spinal, or general anaesthesia, decided by patient, surgeon and anaesthetist.**

- The original MUS description used **LOCAL** anaesthesia specifically so the patient could perform an intraoperative cough stress test to adjust sling tension, on the belief that this would minimise obstruction.
- This belief was not borne out: studies found no difference in efficacy or safety between local and spinal anaesthesia, and the inability to perform a CST under spinal did NOT increase postoperative obstruction. Therefore tensioning to a cough stress test may not improve outcomes.
- Antibiotic prophylaxis, DVT prophylaxis, vaginal preparation and positioning are as for the PVS.
- Additional short-term thigh flexion (≥ 70 degrees) may be beneficial.

Surgical Approach for Retropubic Mid-Urethral Sling

The kit

- Two curved trocars of 5 mm diameter, and a polypropylene mesh segment 40 cm long $\times$ 1.1 cm wide — dimensions vary slightly by manufacturer.
- The sling is covered by a clear plastic sheath, which protects the mesh from contamination and eases passage through host tissues.
- For the down-up (bottom-top) technique, a rigid catheter guide is placed in the urethra with an 18-Fr Foley to deflect the bladder away from the trocar path. An ergonomic handle attaches to the trocar.

Local anaesthetic infiltration

- ~5 mL into the vaginal epithelium and into the planned suprapubic skin sites.
- 20 mL along the posterior aspect of the pubic bone to the level of the urogenital diaphragm, to anaesthetise the retropubic space.
- 10 mL on EITHER side of the urethra to the level of the urogenital diaphragm.

Incisions and dissection

- Two suprapubic stab incisions just above the symphysis pubis, approximately 2 cm lateral to the midline.
- One midline vaginal incision, approximately 1.5 cm long, sited 1.5 cm proximal to the external urethral meatus.
- Minimal dissection with Metzenbaum scissors under the vaginal flaps on either side, elevating vaginal epithelium off periurethral tissue to the level of the pubocervical (endopelvic) fascia.
- **UNLIKE** the PVS, the endopelvic fascia is NOT perforated during dissection.

Bottom-top trocar passage

- Place the trocar in the dissection tunnel immediately beneath the vaginal epithelium on one side of the urethra, tip close to the lower rim of the pubic ramus.
- With controlled pressure, elevate the trocar through the endopelvic fascia $\rightarrow$ into the space of Retzius $\rightarrow$ through the rectus muscles $\rightarrow$ out through the suprapubic skin incision.
- Keep the trocar in close contact with the **INFERIOR SURFACE OF THE PUBIC BONE** throughout, to avoid perforating the lower urinary tract and to avoid intraperitoneal entry.
- Tactile contact with bone plus slow graded pressure ensures metal-to-bone apposition and avoids bladder injury.
- Deflect the lower urinary tract simultaneously with the catheter guide and catheter. Repeat contralaterally.

Top-down trocar passage — the differences

- A catheter guide with Foley is typically NOT used.

- The trocar tip is guided onto the index finger of the opposite hand and out of the vaginal incision lateral to the urethra.
- Trendelenburg may help mobilise abdominal contents cephalad to reduce the chance of visceral puncture.

Cystoscopy

- Performed to exclude trocar penetration of the lower urinary tract.
- A 70-degree lens is **ESSENTIAL**, as is **COMPLETE DISTENTION** of the bladder with irrigant — otherwise subtle tangential injury is missed.
- If perforation is found: withdraw the trocar and repass **MORE LATERALLY**.

Tensioning and closure

- Once cystoscopy is clear, drain the bladder, bring the mesh through the incisions, and tension the sling.
- Tension adjustment: insert a surgical instrument (clamp) or metallic sound between the sling and the urethra while removing the protective plastic sheath.
- The MUS should be placed **LOOSELY** at the mid-urethra — its function is **NOT** primarily related to compression.
- Excise redundant mesh at the level of the suprapubic skin incisions.
- Close the vaginal incision with delayed absorbable suture; skin incisions may be left open or closed by preference.

How surgeons actually tension — Borazjani et al. (2017) survey

- 596 surgeons from 56 countries (roughly two-thirds gynaecologists, one-third urologists), more than 30% performing over 50 slings annually.
- Objective intraoperative tension assessment is uncommon: intraoperative stress test used by a small minority, cystoscopic evaluation of urethral coaptation by fewer, and the cotton-tipped applicator test by very few.
- About two-thirds tension retropubic and transobturator MUS **SIMILARLY**; of the remainder, more placed transobturator slings under greater tension than the reverse.
- Fellowship-trained surgeons were significantly more likely to use leak point pressure results to inform tensioning.

Surgical Approach for Transobturator Outside-In Mid-Urethral Sling

- Position: dorsal lithotomy with legs in **HYPERFLEXION** (120 degrees).
- Vaginal incision as for retropubic: 1.5 cm midline, 1.5 cm from the meatus, but dissection is carried **LATERALLY** to the ischiopubic ramus.
- Skin puncture site: the crease between the labia majora and the thigh, at the notch along the internal edge of the inferior pubic ramus and the adductor longus tendon. Practical landmark: with the patient in hyperflexion, pinch the adductor longus tendon between thumb and index finger — the skin incision corresponds to the position of the **THUMB**.
- Perforate the obturator membrane — resistance is felt at this point, the tactile endpoint.
- Using the non-dominant index finger and the landmarks of the ramus and obturator internus muscle, turn the trocar **MEDIALY**, advance it onto the tip of the index finger, and bring it out through the vaginal incision.
- Inspect to exclude inadvertent penetration of the vaginal fornix ("button hole") or of urinary structures. Repeat contralaterally.

- Cystoscopy after trocar passage is recommended to exclude urethral AND bladder injury — do not neglect careful examination of the URETHRA.
- Attach the sling to the trocar and bring it out through the inner thigh stab wound.
- Tension: a clamp should pass easily between sling and urethra. Cut excess material at the skin puncture site; close incisions by preference.

Surgical Approach for Transobturator Inside-Out Mid-Urethral Sling

- The vaginal component is identical to the outside-in technique.
- Stab incisions: approximately 2 cm SUPERIOR to the horizontal line level with the urethra, and 2 cm LATERAL to the labial folds — the exit point for the helical passer.
- Once dissection reaches the upper part of the ischiopubic ramus, perforate the obturator membrane SHARPLY with scissors.
- Pass the introducer at a 45-degree angle to the midline sagittal plane until it reaches and perforates the obturator membrane; the open side of the introducer faces the surgeon.
- Mount the distal end of the tubing on the spiral segment of the helical passer and slip it along the open gutter of the introducer.
- Align the passer parallel to the sagittal axis and rotate it so the tip of the tubing exits the inner-thigh stab incision.
- Remove the tubing from the passer until the first few centimetres of mesh are externalised; repeat contralaterally.
- Cystoscopy to exclude bladder or urethral injury.
- Remove ALL plastic covering sheaths SIMULTANEOUSLY while maintaining NO tension on the sling.

Surgical Approach for Single-Incision Mid-Urethral Sling

- Each SIMS has a proprietary placement method varying by manufacturer.
- Anaesthesia: parenteral sedation, with approximately 5 mL of 0.9% normal saline to hydrodistend the anterior vaginal wall.
- After the vaginal incision, minimal dissection with Metzenbaum scissors under the vaginal flaps to elevate epithelium off periurethral tissue to the pubocervical fascia — WHICH IS NOT PERFORATED.
- Place one trocar of the device in the RIGHT dissection tunnel immediately beneath the vaginal epithelium, advance to the ischiopubic ramus and INTO THE OBTURATOR INTERNUS MUSCLE, where the trocar holder anchors the sling edge.
- Introduce the left side identically, creating a HAMMOCK-SHAPED sling.
- Definitive tension: the tip of a hemostat passes easily between urethra and sling.
- Cystoscopy with a 30-degree RIGID cystoscope before completing the procedure — essential to avoid the future morbidity of repeat surgery for intravesical or urethral mesh.
- Cumulative experience suggests SIMS should be tensioned TIGHTER than classic retropubic or transobturator MUS to achieve the same result.

OUTCOMES FOR MID-URETHRAL SLING FOR PREDOMINANT STRESS URINARY INCONTINENCE

Caveats in reading the outcome literature

- Outcomes are reported using different tools, follow-up durations and definitions of success and failure.
- Studies may include patients with previous anti-incontinence surgery or concomitant POP repair.

- **Kuprasertkul and Zimmern (2020)**, reviewing long-term data (10–17 years) for the commonest SUI procedures: **all are compromised by considerable loss to follow-up, infrequent examination data, predominantly questionnaire-based information, and variable success rates arising from varying definitions.**

Outcomes for Retropubic Mid-Urethral Sling

- **Ulmsten et al. (1996)**: initial study reported an author-defined success rate of 80%.
- **Ulmsten et al. (1998)**, prospective multicentre trial: 1-year success 91% with a further 7% improved. Complications were few but included **one bladder mesh perforation and one wound infection possibly associated with mesh exposure**;

voiding dysfunction was low, resolving spontaneously or needing only a few days of intermittent catheterisation.

- **The MUS — and the TVT in particular — has become the most studied surgical procedure in the incontinence literature.**
- **Single-centre studies now report follow-up beyond 10 years. Success ranges 48–97% depending on definition, but is typically under 85%.**

Landmark Comparison — Ward and Hilton (2002, 2004, 2008)

- **The benchmark trials comparing retropubic MUS with colposuspension, enrolling the largest numbers and reporting short-, medium- and long-term data.**
- **No difference in cure at 5 years.**
- **MUS: more INTRAOPERATIVE complications (notably bladder trocar injury). Colposuspension: more POSTOPERATIVE complications and longer recovery.**
- **POP was commoner in the Burch group — consistent with earlier studies.**
- **De novo urgency or UUI was low in both: under 2% after MUS and under 5% after colposuspension.**

Retropubic Versus Transobturator — Ford et al. (2017)

- **81 trials, 12,113 females overall; 55 trials with 8,652 participants compared transobturator with retropubic.**
- **Subjective and objective cure are SIMILAR at up to 1 year, and remain similar at medium-term (1–5 years) and long-term (>5 years).**
- **Long-term subjective cure ranges 43–92% transobturator and 51–88% retropubic.**

Where they differ — the examinable trade-off

- **Retropubic MUS carries higher morbidity overall, though absolute adverse event rates remain low.**
- **Favouring TRANSOBTURATOR: markedly lower bladder perforation (0.6% vs 4.5%), fewer major vascular and visceral injuries, shorter operating time, less blood loss, shorter stay, and less postoperative voiding dysfunction.**
- **Favouring RETROPUBIC: less groin pain (6.4% transobturator vs 1.3% retropubic), and limited evidence that repeat incontinence surgery in the long term is more likely after transobturator.**
- **Suprapubic pain is LOWER after transobturator (0.8% vs 2.9%) — the mirror image of groin pain.**
- **Vaginal tape erosion, exposure or extrusion is low and equivalent in both (approximately 2% each).**
- **Within the retropubic approach, the BOTTOM-TOP route is more effective for subjective cure and is associated with significantly lower rates of voiding dysfunction, bladder perforation and vaginal tape exposure.**

Outcomes for Transobturator Mid-Urethral Sling

- Since Delorme (2001), the transobturator MUS has become a viable and effective treatment; studies exceeding 10 years' follow-up are common.
- Overall success in predominantly SUI is similar to retropubic MUS.

Landmark Trial — TOMUS (Richter et al., 2010)

- Trial of Midurethral Slings: prospective randomised controlled trial of retropubic MUS (n = 298) versus transobturator MUS (n = 299), 12 months' follow-up.
- Objective cure — defined by negative cough stress test, negative pad test, and no re-treatment — was 80.8% retropubic versus 77.7% transobturator: NOT statistically different.
- Voiding dysfunction requiring surgery was significantly higher in the retropubic group (2.7% vs 0%).
- Five-year observational cohort (Kenton et al., 2015), enrolling about two-thirds of the original 597: success 51.3% retropubic versus 43.4% transobturator — a difference of just under 8%, which did NOT meet prespecified criteria for equivalence.
- Satisfaction fell over five years but remained high and similar in both arms (79% retropubic, 85% transobturator, p = 0.15).
- AUA SUI Guideline, grade A evidence, Conditional Recommendation: in index patients selecting MUS surgery, physicians may offer EITHER retropubic or transobturator MUS.

Inside-out versus outside-in transobturator

- Short- and medium-term subjective cure are similar.
- Voiding dysfunction is MORE frequent inside-out; vaginal perforation is LESS frequent inside-out.
- Either approach is acceptable, based on surgeon experience.

Outcomes for Single-Incision Mid-Urethral Sling

- First FDA approval 2006: TVT-Secur (Gynecare, Ethicon).

Neuman (2008) — the learning-curve study

- 100 consecutive hammock-style SIMS, comparing the first 50 with the last 50 at 12 months.
- Objective failure fell from 20% to 8%, achieved by placing the sling CLOSER TO THE URETHRA.
- Vaginal perforation with the inserter occurred in the first group and was eliminated in the second by WIDENING THE SUBMUCOSAL TUNNEL.
- Mesh exposure fell from 12% to 8% by creating DEEPER submucosal tunnels.
- No bladder or urethral perforation, UTI or wound infection; one self-remitting paravesical haematoma.

Dmochowski et al. (2009) — one of the largest SIMS trials

- 642 patients across 29 sites in 8 countries, effectiveness by standing cough stress test and the Incontinence Quality-of-Life instrument (I-QOL).
- Sling position: about two-thirds hammock, one-third U-position. Two-thirds pure SUI, one-third stress-predominant MUI.
- Intraoperative complications were rare: bladder perforation and bleeding over 200 mL each well under 1%.
- Postoperative retention, UTI, voiding dysfunction, de novo urgency and mesh erosion were all low (each under 3%).
- Quality-of-life gains at 3 months were sustained at 12 months; objective continence 89% at 6 months and 87.5% at 12 months.

Cochrane Systematic Review (Nambiar et al., 2017)

- **31 randomised or quasi-randomised trials, 3,290 females**, comparing SIMS with other anti-incontinence procedures.
- **SIMS devices included:** TVT-Secur, MiniArc, Ajust, Needleless, Ophira, Tissue Fixation System, CureMesh.
- **No studies compared SIMS against no treatment, conservative treatment, colposuspension, laparoscopic procedures, or traditional PVS.**
- **Findings against SIMS:** more persistent or recurrent incontinence than retropubic MUS (41% vs 26%); higher risk of de novo urgency; higher incontinence than INSIDE-OUT transobturator MUS (30% vs 11%); worse adverse event profile, with higher vaginal exposure and markedly higher bladder and/or urethral erosion.
- **Findings favouring SIMS:** less postoperative pain, with marginally lower long-term pain of questionable clinical significance.
- **The decisive subanalysis:** the excess incontinence was principally attributable to TVT-Secur, which has since been **WITHDRAWN** from the market.
- **With TVT-Secur removed, evidence was INSUFFICIENT** to show any difference in incontinence between other SIMS and inside-out or outside-in transobturator MUS, and insufficient to demonstrate a difference in adverse events. **The only significant differences remaining were lower postoperative and long-term pain, and shorter operating time, marginally favouring SIMS.**
- **Comparing one SIMS against another:** insufficient evidence of any difference between products.

Post-Cochrane meta-analyses

- **SIMS appears NON-INFERIOR** to transobturator MUS on both objective and subjective criteria, at short- and medium-term follow-up.
- **Limited data suggest retropubic MUS outcomes are BETTER than SIMS.** Adverse events after SIMS are low.
- **As with other slings, SIMS efficacy may decline with longer follow-up:** Han et al. (2012) found subjective cure after TVT-Secur fell significantly between years 1 and 3 (85.4% to 72.9%).
- **Masata et al. (2012), randomising SIMS against transobturator MUS at 2 years:** SIMS had significantly lower objective (69% vs 93%) and subjective (75% vs 90%) cure.
- **AUA SUI guidelines, grade B evidence, Conditional Recommendation:** physicians may offer SIMS to index patients undergoing MUS surgery.

Outcomes of Mid-Urethral Sling for Mixed Urinary Incontinence

- **Many patients seeking SUI surgery have some storage LUTS, and their postoperative status is unpredictable.** Storage LUTS after sling surgery is a significant cause of dissatisfaction.
- **Worsened or de novo storage LUTS may represent BOO, which is challenging to distinguish from OAB.**
- **Interpretation is limited because many outcomes are symptom-based with no urodynamic substantiation.**
- **Jain et al. (2011) meta-analysis:** SUI cure after MUS 85–97%; overall subjective cure in MUI was only around 56% at nearly three years; cure of urgency and UUI ranged 30–85% at up to five years, and most studies reported that this cure did NOT persist over time.
- **Shin and Choo (2019), after transobturator MUS:** de novo urgency in a small minority of the pure SUI group; in the SUI-with-urgency group about two-thirds became urgency-free with few developing de novo UUI. UUI resolved in three-quarters of the MUI group overall — **but the gradient by subtype is the**

point: highest resolution in the SUI-PREDOMINANT group (80%), intermediate in the equivalent group, and LOWEST in the UUI-PREDOMINANT group (53%).

- **Whether MUI does worse than pure SUI is contested** — some authors report lower subjective cure in MUI, others similar resolution.
- **Across the literature, resolution of urgency or UUI after MUS is typically 60–75%.**

Perioperative variables influencing MUI outcome

- **Duckett and Basu (2007):** a significantly decreased Qmax after MUS was associated with persistent OAB symptoms; those whose DO resolved had significantly higher Qmax — suggesting **PARTIAL OBSTRUCTION** underlies persistent DO.
- **Gamble et al. (2008):** age, nocturia, maximum capacity and choice of sling procedure affected persistence of DO and UUI. The DO gradient by procedure, lowest to highest: transobturator MUS → retropubic MUS → PVS.
- **Marcelissen and Van Kerrebroeck (2018):** risk factors for de novo urgency or UUI are older age, obesity, parity, nocturia, and OAB-related preoperative urodynamic findings such as DO and low bladder capacity.
- **Panayi et al. (2009):** higher median preoperative **OPENING DETRUSOR PRESSURE** in those with DO after MUS.
- **Natale et al. (2018):** menopause and age over 60 are risk factors for postoperative UUI and overall incontinence in MUI.
- **Botros et al. (2007):** de novo urgency was significantly **LOWER** after transobturator MUS (8%) than after TVT (33%) or SPARC (17%) — though de novo DO and resolution of preoperative DO and UUI did not differ between the three.
- **Paick et al. (2008):** no difference in SUI or overall incontinence cure between transobturator and retropubic; preoperative DO was associated with high UUI persistence in **ALL** groups.
- **Tahseen and Reid (2009),** after transobturator MUS in MUI: UUI persisted in about a fifth, improved in about a third, and resolved in about two-fifths.

Summary

- **Cure is typically higher in pure SUI than in MUI, but MUS is an effective treatment where urgency or UUI coexists.**
- **Over half of patients with preoperative MUI experience improvement or resolution of urgency or UUI.**
- **Urge-predominant MUI is more likely to require COMBINATION therapy addressing both stress and urge components.**
- **De novo urgency and UUI may develop after ANY sling procedure.**

Outcomes of Mid-Urethral Sling for Intrinsic Sphincter Deficiency

- **ISD is an important risk factor for failure of conventional anti-incontinence procedures, but the LACK OF A UNIVERSALLY ACCEPTED DEFINITION hinders evaluation of surgical efficacy.**
- **Urodynamic definition as before: ALPP <60 cm H₂O or MUCP <20 cm H₂O.**

Evidence that ISD predicts failure

- **Lo et al. (2022),** in over 600 MUS patients of whom 8% had urodynamic ISD: at one year, **ISD patients had markedly lower objective and subjective cure (around 50%) versus overall cure of around 86–88%. Sling type made no difference,** and on multivariate analysis **reduced urethral mobility and lower MUCP were associated with higher odds of failure.**

- **Paick et al. (2004):** cure was significantly lower with VLPP under 60 cm H₂O than with higher VLPP (82.0% vs 93.1%, $p = 0.013$).

The counter-argument — mobility, not pressure

- Several authors hold that **LOSS OF URETHRAL MOBILITY**, rather than low ALPP or MUCP, is the true risk factor for failure.
- Therefore low leak point pressures are NOT in themselves a contraindication to retropubic MUS; in the MUS era, leak point pressures should be correlated with the physical examination and used to counsel about reduced chance of success.
- Success is clearly lower in those with BOTH a fixed urethra and low leak point pressures.
- A preoperatively **FIXED URETHRA** is more predictive of no cure than a low leak point pressure.

Which sling in ISD

- Ample evidence supports retropubic MUS in ISD.
- Evidence also exists for non-inferior transobturator MUS outcomes in ISD — but O'Connor et al. (2006) found the odds of continued SUI after transobturator MUS were TWELVE TIMES greater with low VLPP.
- Comparative studies and meta-analyses indicate **RETROPUBIC MUS IS MORE EFFECTIVE THAN TRANSOBTURATOR MUS IN FEMALES WITH ISD**.
- TOMUS, though not designed for this question, performed preoperative urodynamics on all patients and found VLPP and MUCP had NO effect on sling outcomes — but urethral mobility was not recorded, which is precisely the omitted variable.
- Limited evidence suggests SIMS performs similarly to transobturator MUS in ISD (Hwang et al., 2022).
- MUS is beneficial in SUI with ISD PROVIDED there is preoperative urethral mobility, and absolute cure and improvement rates in ISD fall within the range reported for other procedures.
- AUA/SUFU Expert Opinion: in patients with SUI and a **FIXED, IMMOBILE** urethra who wish treatment, physicians may offer retropubic MUS, PVS, bulking, or adjustable retropubic MUS.

Outcomes of Mid-Urethral Sling for Recurrent Stress Urinary Incontinence

- Recurrent or persistent SUI after failed anti-incontinence surgery is a genuine dilemma: there is NO level 1 evidence defining the best next step after a failed MUS.
- A Cochrane review concluded there were insufficient data to assess the effects of ANY management strategy (Bakali et al., 2019).
- No published paper reports exclusively on females whose FIRST operation was a MUS — the population of interest has never been studied in isolation.
- British Society of Urogynaecologists database review of 2,938 records: repeat MUS is by far the commonest procedure performed (77.3%) (El-Hamamsy and Tincello, 2021).
- Comparison across studies is hampered by differing definitions of cure and improvement, differing outcome criteria, and short follow-up; many series are further biased by including patients in whom bulking agents had failed.

Choice of procedure

- Retropubic MUS performs BETTER than transobturator MUS for recurrent SUI.
- Sabadell et al. (2020): retropubic MUS after transobturator MUS failure gave a cured or improved rate of 67% at 10 years.
- Abdel-Fattah et al. (2017b): transobturator MUS for recurrent SUI gave 62% success by Patient Global Impression of Improvement at 9 years, with clinically significant quality-of-life improvement in 84%.

- **Agur et al. (2013)**, meta-analysis of 10 randomised trials: **NO significant difference in subjective cure between retropubic and transobturator.**
- **Kim et al. (2019a)**, reviewing 28 studies and 2,607 patients: subjective cure after the **RETROPUBIC** procedure was **SIGNIFICANTLY SUPERIOR** in recurrent SUI after previous MUS — the more recent and larger analysis.

Repeat versus primary sling

- **Stav et al. (2010a)**: repeat MUS was significantly **LESS** effective at curing incontinence than a primary sling (**62% vs 86%**). Overall complication rates were similar, but the **repeat group had significantly more de novo urgency or UUI.**
- **Confounder to note:** the repeat group had significantly more preoperative ISD (**31% vs 13%**) — so the inferiority may reflect the population rather than the operation.

Cure rates and practical trends

- **MUS after prior failed anti-incontinence surgery generally: cure 81–90%.**
- The procedure can be performed in the same way as for primary SUI.
- **Complication rates resemble those of primary retropubic MUS — EXCEPT that bladder perforation risk is higher after one or more prior retropubic suspensions.**
- As in primary surgery, failure is higher in patients with **IMMOBILE** urethrae.

KEY POINTS: MID-URETHRAL SLINGS FOR RECURRENT STRESS URINARY INCONTINENCE

- A repeat mid-urethral synthetic sling for persistent or recurrent SUI is a viable option in **SELECTED** patients in whom the initial procedure failed.
- Some studies show higher cure with retropubic than transobturator slings — possibly the result of unequal rates of ISD across study populations.

Outcomes of Mid-Urethral Sling in Patients With Pelvic Organ Prolapse

The three possible continence states in prolapse

- **Overt SUI** — subjective and objective, symptomatic.
- **Occult SUI** — demonstrated only with POP reduction.
- **Neither.**
- **Critical caveat:** absence of symptomatic, overt or occult SUI may mean there is no SUI — or may simply reflect the **LIMITATION** of demonstrating SUI with prolapse reduction. Rarely, the patient may misinterpret SUI as UUI.

The concurrent-versus-staged dilemma

- **Concurrent sling:** reduces postoperative symptomatic SUI, but increases sling-related complications — obstruction, de novo or worsened storage LUTS, and mesh-related adverse events.
- **Staged sling:** treats SUI only if it develops and minimises sling complications during the POP repair — but commits the patient to a second operation if SUI does occur.
- **Giugale et al. (2019):** preoperative SUI **RESOLVED** in about 30% after POP repair alone; about half were treated for persistent SUI, and **about a third underwent a staged MUS, most within 12 months.**

Evidence supporting a concomitant sling for symptomatic or occult SUI

- **Baessler et al. (2018)**, meta-analysis of 19 randomised trials and 2,717 females: a **concomitant MUS** probably improves postoperative subjective SUI rates and probably decreases the need for further continence surgery. Illustratively: if the risk of SUI after POP surgery alone is 39%, the risk with a MUS falls to between 8% and 19%.

- **van der Ploeg et al. (2016)**, randomising occult SUI to POP surgery alone versus combined: **13% needed a staged MUS in the POP-alone arm, versus none in the combined arm.**
- **van der Ploeg et al. (2018)**, meta-analysis of 10 trials: **combination surgery reduces postoperative SUI, but short-term voiding difficulties and adverse events are MORE frequent.**
- **Maher et al. (2013)**, Cochrane analysis of 16 studies: **patients undergoing POP repair may benefit from concomitant anti-incontinence surgery, ESPECIALLY with overt SUI, or if continent with occult SUI demonstrated preoperatively. After POP repair, de novo urgency and UI occurred in 12% and de novo voiding dysfunction in 9%.**
- **Anger et al. (2008)**, Medicare data, about a third with concomitant POP repair: **significantly more postoperative BOO (9.4% vs 5.5%), but LESS reoperation for SUI or POP within one year.**

Theoretical advantages and risks of a MUS at prolapse repair

- **Advantage over autologous PVS or retropubic suspension: reduced operative time and blood loss from the sling portion.**
- **Increased dissection increases graft exposure → greater infection, perforation or vaginal exposure.**
- **Increased blood loss or anatomical distortion from the POP procedure → sling migration, voiding dysfunction, obstruction.**
- **Appropriate TENSIONING is challenging in occult SUI, where the leak is only demonstrable with the prolapse reduced.**

Outcomes in practice

- **Studies are mostly small and short-term, but MUS can be added to POP surgery with minimal morbidity and high success (73–93%) — and the same applies to MUS for occult SUI.**
- **Meltomaa et al. (2004)**, at three years by questionnaire: **no statistical difference in SUI cure or urgency between MUS alone and MUS combined with other vaginal surgery. Transient retention was commoner after combined surgery, but urethrolisis rates were low and not statistically different.**
- **Bakas et al. (2018)** at 13 years: **anterior colporrhaphy cured POP in 92%, and cured cystocele with urodynamic SUI in 84% when combined with TVT-O; TVT-O alone gave objective cure of 81%.**
- **AUA/SUFU Conditional Recommendation, grade C evidence: in patients undergoing concomitant surgery for POP and SUI, physicians may perform ANY of the incontinence procedures — MUS, PVS, or Burch colposuspension.**

The "prophylactic" sling — prolapse with NO symptomatic, overt or occult SUI

- **Stewart et al. (2021)**, comparing MUS for documented SUI, prophylactic MUS, and no MUS: **no significant difference in postoperative retention in those without a sling; persistent SUI in 7% of the documented-SUI group, de novo SUI in 5% of the prophylactic group, and de novo SUI in 12% of the no-MUS group.**
- **Lo et al. (2019)**, hysterectomy and transvaginal mesh with MUS versus POP repair without MUS: **de novo SUI 5% objective and 10% subjective WITH a sling, versus 50% objective and 60% subjective WITHOUT.**
- **Matsuoka et al. (2015)**, systematic review of seven randomised trials: **an anti-incontinence procedure at the time of POP repair reduced postoperative SUI — but on subgroup analysis, ONLY the RETROPUBIC MUS subgroup derived benefit.**

Landmark Trial — OPUS (Wei et al., 2012)

- **Outcomes Following Vaginal Prolapse Repair and Midurethral Sling. Question: does a concomitant "prophylactic" RETROPUBIC MUS prevent SUI in patients undergoing POP surgery with NO clinical (subjective) SUI preoperatively?**
- **Design: sham incisions in the control arm; 327 patients without clinical incontinence completed 1-year follow-up.**

- **Result: urinary incontinence at 12 months (allowing for subsequent treatment) in 27% of the sling group versus 43% of the sham group, $p = 0.002$.**
- **Number needed to treat with a sling to prevent one case of SUI at 12 months: 6.3.**
- **Trocar bladder injury, UTI and bleeding were all low but SIGNIFICANTLY HIGHER in the MUS group.**
- **Important interpretive caveat: some in the sham group, though without subjective SUI, may have had OCCULT SUI, inflating the failure rate in that arm.**
- **Author's opinion: a TRUE prophylactic MUS — in a patient with no subjective, overt, occult or urodynamic SUI — should be placed only after extensive and thorough preoperative counselling regarding expectations and goals.**

Outcomes of Mid-Urethral Sling in Elderly Females

Why age matters

- **Higher rates of DO, UUI, ISD, and emptying abnormalities from impaired contractility.**
- **More likely to have had prior incontinence procedures → higher rates of URETHRAL FIXATION, itself a predictor of sling failure.**
- **Long-standing oestrogen deficiency → greater risk of poor or delayed healing and mesh exposure.**
- **Greater anaesthetic risk from medical comorbidity, affecting the surgery, its outcome and the postoperative course.**

Gordon et al. (2005) — 460 consecutive MUS, about a third aged over 70

- **Significantly FEWER bladder trocar injuries in the elderly.**
- **Postoperative fever, UTI, wound infection and haematoma were similar between groups.**
- **More perioperative medical morbidity in the elderly — pulmonary embolism, cardiac arrhythmia, DVT and pneumonia — against a single arrhythmia in the younger group.**
- **NO increased risk of vaginal exposure. Voiding dysfunction requiring catheterisation beyond one week was low and similar in both groups.**
- **Cure rates essentially identical: 93% elderly versus 94% younger. De novo UUI was significantly greater in the older patients (18% vs 4%).**

Other de novo urgency data

- **Sevestre et al. (2003): a similar rate of de novo UUI (around 18%) in elderly patients.**
- **Allahdin et al. (2004): postoperative urgency in as many as 60% of older females, of whom 44% developed it de novo — the highest figures reported.**
- **Athanasίου et al. (2020), 72 patients aged over 75, mean follow-up 13.7 years after TVT-O: objective cure 80.5%, subjective 84.7%, patient-reported success 91.7%. De novo urgency 23.7%; aggravation of pre-existing urgency in a further 26.5% — distinct endpoints, both to be counted.**

Urethral mobility as the decisive variable — Liapis et al. (2006)

- **Cotton-tipped applicator displacement <30 degrees: only 42% became continent.**
- **Displacement >30 degrees: 90% continent.**
- **Angle <10 degrees: 80% remained INCONTINENT.**

Cure data and health burden

- **Jha et al. (2009): age did NOT significantly affect SUI reduction or quality-of-life improvement after retropubic MUS.**
- **Postoperative voiding dysfunction and de novo urgency do appear commoner in older females, and may affect them more significantly.**

- **Bafghi et al. (2005):** significantly higher cure (98% vs 79%) and satisfaction (93% vs 67%) in younger patients — attributed to higher rates of de novo and persistent UUI in the older group.
- **Hellberg et al. (2007):** significantly lower cure in older patients (56% vs 80%) — but attributed to a higher SUI FAILURE rate, NOT to urgency symptoms.
- **Gyhagen et al. (2023),** 5,200 patients aged 55–94 from the Swedish National Quality Register of Gynecological Surgery: cure 64% in the ≥75 cohort versus 89% in the 55–64 cohort ($p < 0.0001$). ASA class 3–4 patients had lower cure (66% vs 84%) and lower satisfaction (66% vs 81%), both $p < 0.0001$. Older age was associated with de novo urgency and with nocturia of ≥ 2 episodes. Adverse events, readmissions and 30-day mortality were all low.
- **The key patient-centred finding:** irrespective of age, patients were **EQUALLY SATISFIED** if they experienced a decrease of at least one step in leakage frequency. Despite lower cure, the majority of elderly patients and those with diminished physical status were satisfied.

Frailty rather than chronological age

- **Emphasis has shifted from chronological age to FRAILITY** as the risk factor for surgical complications. Full discussion in Chapter 120.
- **Medendorp et al. (2021),** from over 25,000 sling procedures in the National Surgical Quality Improvement Project, of whom 10% were frail: adverse events were low overall; **frailty was associated with increased length of stay and increased 30-day readmission; older age alone predicted longer operating time and longer stay after isolated sling placement.**
- **The decisive comparison:** in sling with concomitant POP repair, **FRAILITY** predicted 30-day readmission whereas **AGE** did NOT.

Summary and guideline

- **Persistent SUI after retropubic or transobturator MUS is higher in the elderly,** potentially due to concomitant UUI or reduced urethral mobility. The elderly also have a higher incidence of de novo urgency and UUI.
- **Despite some initial increase in perioperative morbidity, MUS surgery IS safe in the elderly.**
- **The literature is insufficient to support one anti-incontinence procedure over another in this group.**
- **Elderly patients should NOT be excluded from potentially curative MUS surgery on the basis of age alone.**
- **AUA/SUFU Expert Opinion:** physicians may offer synthetic MUS to the elderly after appropriate evaluation and counselling.

Outcomes of Mid-Urethral Sling in Obese Patients

- **The prevailing link between obesity and SUI is INCREASED INTRA-ABDOMINAL PRESSURE.** Neurological and metabolic mechanisms have also been proposed, supporting a multifactorial pathophysiology.
- **Both retropubic and transobturator MUS achieve success exceeding 80% in obese females, varying with the definition of cure.**
- **Success is nonetheless LOWER than in the non-obese, and several long-term observational studies identify OBESITY AS A RISK FACTOR FOR TVT FAILURE.** Brennd et al. (2017), 5-year secondary analysis of a randomised trial: objective cure by 1-hour pad test 87.4% non-obese versus 65.9% obese ($p = 0.003$); subjective cure 76.7% versus 53.6% ($p = 0.002$).
- **SIMS data in obesity are CONFLICTING** — one study found worse outcomes in the obese, another did not. **Both are methodologically compromised:** the negative study used TVT-Secur in nearly half its patients, and the positive study used a BMI cut-off of 25 kg/m², thereby grouping overweight women with the obese.

- **Outcomes in the MORBIDLY obese (BMI >35 kg/m²) appear lower than in the overweight and obese population.**

Complications — the data largely refute the expected excess

- **Weltz et al. (2015)**, systematic review of pooled data from 13 studies: **no significant difference in urinary retention or sling excision between obese and non-obese.**
- **Chen et al. (2007)**: **no statistical difference in the proportion having at least one perioperative complication (20% obese vs 15% non-obese) after vaginal surgery. Obese patients WERE significantly more likely to have an operative site infection (adjusted odds ratio 5.5, p = 0.01).**
- **Bladder puncture during retropubic MUS appears LESS common in the obese.**
- **Zyczynski et al. (2015)**: **BMI ≥30 kg/m² reduced the likelihood of OAB symptom improvement after SUI surgery.**
- **Cadish et al. (2017)**: **after adjusting for age and activity level, overweight and obese patients had significantly increased risk of postoperative-onset PAIN after inside-out transobturator MUS.**

TABLE 118.5 OUTCOMES WITH RETROPUBIC MID-URETHRAL SLINGS (>10 YEARS OF FOLLOW-UP)

- **Columns:** lead author and year · number · type · follow-up (years) · cured · assessment of outcome · de novo urgency (%) · any urinary retention (%).
- **The overwhelming majority of entries are TVT**, with occasional other devices — **DUPS (distal urethral polypropylene sling)** and unspecified MUS.
- **Follow-up extends to 17 and even 20 years** — the longest published experience of any anti-incontinence procedure.
- **Objective cure is consistently reported HIGHER than subjective cure within the same series**, commonly by 10 to 25 percentage points. This gap is the single most important reading of the table.
- **Objective cure remains in the high eighties to low nineties even at 17 and 20 years** in several series, supporting genuine durability.
- **Subjective cure is far more variable, from about 40% to the mid-nineties**, tracking the instrument used rather than the operation.
- **Where a Patient Global Impression of Improvement (PGI-I) score is reported, it is HIGHER than either objective or subjective cure.**
- **De novo urgency rises with follow-up duration**, reaching around a third to over 40% in the longest series; **any urinary retention is usually low (0–7%)**, with occasional outlying series reporting substantially more.
- **Key:** CST, cough stress test · DUPS, distal urethral polypropylene sling · MUS, mid-urethral sling · NR, not recorded · Obj, objective · PGI-I, Patient Global Impression of Improvement Scale · RP, retropubic · Subj, subjective · TVT, tension-free vaginal tape · UD, urodynamic · VQ, validated questionnaire.

TABLE 118.6 LONG-TERM OUTCOMES WITH TRANSOBTURATOR MID-URETHRAL SLINGS (>8 YEARS)

- **Columns:** lead author and year · number · type · follow-up (years) · cured · assessment of outcome · de novo urgency (%) · any urinary retention (%).
- **Devices named: TOT (transobturator tape), TVT-O, TOM, Monarc, Aris.** TVT-O denotes the inside-out route; TOT conventionally the outside-in.
- **Note the shorter inclusion threshold: >8 years, against >10 years for retropubic** — reflecting the transobturator's later introduction.
- **Long-term objective cure spans a wide range, from around 57% to the low nineties**, broadly overlapping the retropubic figures but with a lower floor.

- **The objective-versus-subjective gap persists, and in several series runs in BOTH directions** — some report subjective cure exceeding objective, unlike the retropubic table.
- **Several series distinguish de novo OAB from de novo UII, and worsened urgency from new urgency** — categories worth noting as distinct endpoints.
- **Retention and de novo voiding dysfunction are reported separately in several series**, with voiding symptoms sometimes substantially commoner than frank retention.
- **Key:** CST, cough stress test · NR, not recorded · OAB, overactive bladder · Obj, objective · PGI-I, Patient Global Impression of Improvement · Subj, subjective · TOM and TOT, transobturator tape · TVT-O, tension-free vaginal tape obturator · UD, urodynamic · VQ, validated questionnaire.

TABLE 118.7 LONG-TERM OUTCOMES WITH SINGLE-INCISION MINI SLINGS (>9 YEARS)

- **Columns:** lead author and year · number · type · follow-up (years) · cured · assessment of outcome · de novo urgency · REOPERATION (%).
- **Note the changed final column: REOPERATION, not retention** — reflecting that the governing long-term concern with SIMS is **the need for further surgery**, not obstruction.
- **Devices: TVT-S (TVT-Secur) and MiniArc. The table is short — only three series** — confirming that long-term SIMS data remain sparse compared with the other two routes.
- **Long-term cure is reported around 58–88%**, with the **withdrawn TVT-Secur at the lower end** and MiniArc higher.
- **Reoperation is separated into reoperation FOR SUI and reoperation FOR COMPLICATION** — the former substantially commoner. **De novo urgency, where recorded, is high (around a third).**
- **Key:** CST, cough stress test · NR, not reported · Obj, objective · PGII, Patient Global Impression of Incontinence Scale · Subj, subjective · SUI, stress urinary incontinence · VQ, validated questionnaire.

TABLE 118.8 META-ANALYSES COMPARING SINGLE-INCISION MINI SLINGS TO OTHER ANTI-INCONTINENCE PROCEDURES

- **Columns:** lead author and year · number of randomised trials and females · SIMS device or devices · comparator MUS · summary of findings.
- **Comparators are almost exclusively TOT/TVT-O**, with occasional retropubic MUS. **SIMS devices assessed:** Ajut, MiniArc, Contasure-Needleless (C-NDL), Ophira, Tissue Fixation System (TFS).
- **SIMS is NON-INFERIOR to transobturator MUS for SUBJECTIVE cure in every analysis. For OBJECTIVE cure, most find non-inferiority, but at least one found transobturator MUS SUPERIOR** — the one point of genuine disagreement.
- **SIMS advantages are consistent and perioperative: shorter operating time, less estimated blood loss, less immediate postoperative pain, less postoperative groin pain, and in some analyses less voiding dysfunction**

and less retention. Overall adverse event profiles are not statistically different.

- **Limited comparisons suggest RETROPUBIC MUS is more effective than SIMS. SIMS versus SIMS comparisons remain inconclusive;** no product is established as superior.
- **The network meta-analysis compares individual devices across many endpoints** — subjective and objective cure, operating time, blood loss, postoperative stay, groin pain, retention, repeat surgery, erosion, urinary infection, de novo urgency, and sexual intercourse. **No single device ranks best across all domains;** the leaders differ endpoint by endpoint, which is itself the finding. **TVT-O consistently ranks POOREST for the perioperative endpoints.**

- **Footnotes:** one analysis included both randomised trials and retrospective cohort studies; in another, several studies had a combination of two SIMS or two MUS in a single trial arm, or compared two separate SIMS individually against one MUS. **"Long-term" here is defined as 5 years.**
- **Key:** AE, adverse event · C-NDL, Contasure-Needleless · EBL, estimated blood loss · LT, long-term (5 years) · MUS, mid-urethral sling · Obj, objective · RP, retropubic · SIMS, single-incision mini sling · Subj, subjective · TFS, Tissue Fixation System · TOT, transobturator tape · TVT-O, tension-free vaginal tape obturator.

KEY POINTS: MID-URETHRAL SLING OUTCOMES IN FEMALES WITH CONCOMITANT PELVIC ORGAN PROLAPSE, ADVANCED AGE, AND OBESITY

- The literature supports MUS across a variety of special populations: efficacy and safety are NOT compromised in the elderly, the obese, or those undergoing concomitant vaginal surgery.
- A few small studies suggest concomitant vaginal surgery raises the risk of delayed return to normal voiding and of urinary retention — but study design limitations prevent strict comparison.
- Current sling data are difficult to interpret because of the absence of standard definitions of cure and improvement and the absence of standard outcome measures.
- Absolute conclusions about MUS in special populations should not be drawn until larger prospective multicentre studies are performed.

COMPLICATIONS OF RETROPUBIC, TRANSOBTURATOR, AND SINGLE-INCISION MID-URETHRAL SLING

The regulatory context

- Over the past decade there has been a 27% increase in the rate of surgery for SUI, largely attributable to the growing popularity of the MUS (Jonsson Funk et al., 2012).
- 2010: the FDA released a revised statement alerting the public to safety concerns about synthetic mesh for POP and SUI surgery.
- The FDA became aware of the problem through the Manufacturer and User Facility Device Experience (MAUDE) database.
- MAUDE data, 2008–2010: 1,371 voluntary and involuntary self-reported medical device reports of MUS-related complications, a significant proportion being mesh perforation and exposure.
- Full discussion of mesh complications, and of the regulatory and legal issues, is in Chapter 119.

Mid-Urethral Sling Trocar Injury to the Urinary Tract

- Trocar injury = passage of the trocar into the urethra or bladder at the time of sling placement. This is **DISTINCT** from perforation and exposure, which are late events discussed in Chapter 119.
- Postoperative quality of life may be affected as much by voiding difficulties, UTIs and other adverse events as by cure or improvement of SUI.
- Causes: a combination of patient and technical factors — body habitus, previous pelvic or vaginal surgery, iatrogenic injury, and surgeon technical error.

Incidence

- Early national registries soon after TVT introduction: bladder trocar injury 2.7% (Finland) and 3.8% (Austria) — consistent with current literature.
- Crane et al. (2023), 1,500 MUS from a single centre: 2.3% trocar puncture rate, ALL in retropubic cases. At 34 months, no significant difference in de novo storage or emptying symptoms between punctured

and non-punctured patients, and cystoscopy in the punctured group showed **NO** bladder mesh exposure.

- **Ford et al. (2017):** bladder puncture 4.5% retropubic versus 0.6% transobturator — the rate is lower with transobturator **because of the direction of trocar passage.**
- **Within the transobturator approach, injuries may be commoner with the OUTSIDE-IN route** (Tamussino et al., 2007).

Management — the critical asymmetry

- **BLADDER** trocar injury is generally **BENIGN**. No study has linked trocar injury to perioperative haemorrhage, haematoma, or subsequent mesh perforation of bladder or urethra.
- Once recognised intraoperatively: re-pass the trocar safely with cystoscopic confirmation, and **CONTINUE** the case. An indwelling catheter may be left overnight or up to 72 hours, though some describe removal in the recovery room with no untoward sequelae.
- **URETHRAL** trocar injury is a **CONTRAINDICATION** to continuing MUS surgery. AUA/SUFU Clinical Principle: physicians should **NOT** place a mesh sling if the urethra is inadvertently injured at the time of planned MUS.
- Re-evaluate for **STAGED** anti-incontinence surgery after catheter drainage and complete healing of the urethral puncture.

Other Trocar Injuries and Bleeding After Mid-Urethral Sling

- Serious injuries are extremely rare: the Finnish registry reported serious vascular complications in 0.07%, the Austrian registry bowel perforation in 0.04%.
- Deng et al. (2007), reviewing MAUDE: of 928 reported complications, 161 (17.3%) were considered **MAJOR** — including vascular injuries, bowel injuries, and 10 deaths.
- Bleeding is inconsistently reported. Large registries and randomised trials: severe bleeding or haematoma in approximately 2–3%, usually manageable with observation or local compression.
- Lin et al. (2021), systematic review of 55 studies: haemorrhage rates of about 1.6% retropubic and 1.5% transobturator.
- Bleeding complications occur more commonly after **RETROPUBIC MUS**: Flock et al. (2004) reported haemorrhage of ≥ 300 mL requiring surgical intervention in about 4%.
- Tseng et al. (2005), performing ultrasonography on every patient after MUS: **about an eighth had significant retropubic haematomas over 5 cm on the first postoperative day; only one required evacuation, and all others had resolved by one month.**
- The important conceptual point: a postoperative haematoma is a relatively **COMMON** and **ASYMPTOMATIC** event after trocar passage — its detection does not mandate intervention.

Urinary Tract Infections

- UTI is arguably the commonest complication after MUS surgery.
- Bonus et al. (2022), national surgical quality analysis of 16,491 MUS: **UTI was the commonest complication (3.4%).**
- Dallas et al. (2017), from Californian statewide data, 28,635 outpatient MUS as a sole procedure: **within 30 days, 5.7% had at least one unplanned hospital visit; among emergency department diagnoses, UTI was second only to urinary retention and catheter problems.**
- Berger et al. (2022), 13,404 MUS procedures from a large managed care organisation: **in the first 12 postoperative months, 23% developed a UTI and 4% developed de novo recurrent UTI.**

- **Timing:** UTIs were most frequently diagnosed in the **FIRST MONTH**, with the **SEVENTH POSTOPERATIVE DAY** the commonest single day.
- **Predictors:** increased age (the strongest), diabetes, and urinary retention requiring prolonged catheterisation. UTI was **LESS** likely after transobturator MUS.
- **The consequential finding:** patients who developed a UTI were significantly more likely to undergo **REOPERATION** — including mesh revision or removal, and surgery for recurrent SUI — and significantly more likely to develop postoperative de novo UUI. UTI is therefore a marker of a worse overall trajectory.
- **Vigil et al. (2017)**, reviewing 9,022 MUS procedures: **age over 65**, **BMI over 40 kg/m²**, and **hospital admission** were independently associated with increased infection risk.

Groin and Thigh Pain After Mid-Urethral Sling

- Groin and thigh pain are associated with the **TRANSOBTURATOR** approach; suprapubic pain with the **RETROPUBIC** approach.
- **Laurikainen et al. (2007)**, Finnish randomised trial: groin pain in 16% after inside-out transobturator versus 1.5% after retropubic MUS.
- **Teo et al. (2011)**: a randomised trial comparing the two routes was **ABORTED** because of a 26% incidence of leg pain in the transobturator arm (versus 1.7% retropubic).
- **Ford et al. (2017)**, from 18 trials and 3,221 females: pooled groin pain after transobturator MUS 6.4%. Groin pain may also **PERSIST** longer after transobturator MUS.
- **Doo et al. (2006)**: persistent suprapubic pain in about 2% at 5 years after retropubic MUS.
- The majority of groin pain is **TEMPORARY**. **De Leval (2003)**: the symptom, present in 16%, resolved after the **SECOND** postoperative day.
- **Krauth et al. (2005)**: most affected patients responded to non-steroidal anti-inflammatory drugs. Proposed mechanisms: subclinical haematoma, or a transient neuropathic phenomenon.
- **Lim et al. (2006)**: early groin pain in about a quarter after transobturator MUS, but persistent groin pain at 12 months in only 3.7% after conservative management — the trajectory to counsel.
- **Roth (2007)** and **Duckett and Jain (2005)**: steroids and local anaesthetic effective, without side effects, for persistent groin pain at 3 months. Rest and physical therapy may also help.
- The extreme case: **Wolter et al. (2008)** reported refractory medial thigh pain requiring medial thigh and transobturator **EXPLORATION** by orthopaedic surgery, with sling excision.

Wound Infections After Mid-Urethral Sling

- Wound-related complications range from minor superficial cutaneous infection to pelvic abscess.
- **Ward and Hilton (2002)**: vaginal wound infection in 2% after retropubic MUS. **Richter et al. (2010)**: 0.7% in **BOTH** retropubic and transobturator arms.
- Rare cases of **NECROTISING FASCIITIS** have been reported, associated with obesity, hypertension and diabetes.
- Severe infection is rare, but diagnosis requires a **HIGH INDEX OF SUSPICION** and may be **DELAYED BY UP TO SEVERAL YEARS**. It may be associated with mesh exposure; treatment is discussed in Chapter 119.
- Abscess and **ADDUCTOR MYOSITIS** have been reported, presenting as leg pain, difficulty ambulating, and cellulitis.

Mesh excision for chronic pain and severe infection

- Where chronic pain or severe infection follows a mesh MUS and non-operative therapy fails, complete mesh excision **FROM BOTH SIDES OF THE BONE** may be necessary.

- **Retropubic mesh → abdominal AND vaginal incisions. Transobturator mesh → medial thigh AND vaginal incisions.**
- **Zeng et al. (2022)**, 87 patients undergoing synthetic MUS removal for pain (complete in 49, partial in 38): pain improved or resolved overall in 78.1%, and **COMPLETE** removal gave a significantly higher chance of pain **RESOLUTION** (63.3%) than partial removal (26.3%), $p = 0.002$ — regardless of sling type. Overall complication rate was low (5.7%), mostly transfusions.

Emptying Dysfunction After Mid-Urethral Sling

- Voiding symptoms — slow stream, hesitancy, intermittency, frank retention — are typically the result of BOO from the sling being placed too **TIGHTLY**, in the **WRONG LOCATION** (too **PROXIMALLY**), or associated with POP that was unrecognised preoperatively or arose de novo.
- Some patients have voiding dysfunction **WITHOUT** evidence of obstruction.
- Originally all slings were thought to carry a similar risk; it is now believed the **RETROPUBIC MUS** confers a **HIGHER** risk.
- **TOMUS**: voiding dysfunction necessitating surgery or permanent catheterisation 2.7% retropubic versus 0% transobturator. Urinary retention, defined as catheterisation beyond 6 weeks, was also higher retropubic (3.7% vs 0.7%).
- **Ford et al. (2017)**, from 37 trials and 6,200 females: postoperative voiding dysfunction significantly **LESS** frequent after transobturator MUS.
- Small studies have associated postoperative voiding dysfunction with **preoperative Qmax and PdetQmax**, addition of other pelvic surgery, increasing age, **DECREASING BMI**, postoperative UTI, and previous anti-incontinence surgery — but there is **NO** consensus on preoperative factors predicting voiding dysfunction after MUS.

Evaluation

- The optimal evaluation is poorly defined in the literature.
- The decision to perform urethrolisis or sling incision rests on a **CLEAR TEMPORAL RELATIONSHIP** between symptom onset and the operation.
- Careful pelvic examination must exclude progression of **ANTERIOR AND APICAL POP**, which can render a non-obstructive sling obstructive **TEN YEARS OR MORE** after placement.
- Urodynamics are useful in selected cases at the physician's discretion — but **NOT** all patients will show obstruction urodynamically.
- Cystoscopy is useful to exclude bladder pathology, urethral mesh perforation, and a hypersuspended bladder neck.

Management — conservative first

- **Obstruction after MUS is FREQUENTLY TRANSIENT** and can be managed with short-term CIC or continuous catheter drainage.
- If voiding volitionally but inefficiently: timed voiding, biofeedback, PFMT and selective medical therapy have all had some success.
- Finnish database (Kuuva and Nilsson, 2002, 2003): of 34 patients with urinary retention, most (20) resumed normal voiding after fewer than 14 days of conservative management; two took a further 5 and 6 weeks; only **ONE** required midline sling lysis, after which voiding was normal. Separately, 111 patients had voiding dysfunction **WITHOUT** retention: in 13 it lasted up to 4 months, and only 2 required surgical transection of the sling.
- Long-term retention after retropubic MUS is **RARE**; incision, revision or removal of the sling usually improves symptoms.

- Some reports show benefit from urethral dilation or loosening the sling under anaesthesia, and early sling loosening through a TRANSVAGINAL approach is a reasonable course of action.
- The author's position: urethral dilation is at best of LIMITED usefulness and at worst DETRIMENTAL, because of its potentially traumatic nature — it may induce urethral scarring or lead to mesh perforation. The author advocates STRONGLY AGAINST urethral dilation in this setting.

Sling incision — technique and timing

- Within the first 3 months: midline division of the MUS through a SINGLE VERTICAL VAGINAL INCISION is the preferred method for persistent obstructive voiding dysfunction.
- After 3 months the sling is fixed along its ENTIRE COURSE, and midline incision may not achieve enough relaxation to resolve the dysfunction. More extensive sling excision is then required, including the LATERAL portion — extending to the inferior portion of the pubic symphysis for retropubic MUS, and to the ischiopubic ramus for transobturator MUS.
- Rarely, a full urethrolisis as described for the autologous BNS is required.
- Author's observation from hundreds of revisions: OBSTRUCTIVE SLINGS ARE MOST OFTEN FOUND CLOSER TO THE BLADDER NECK THAN THE MID-URETHRA — malposition, too proximal, is the usual finding.
- Most studies report 70–100% remain continent after sling lysis, though one large series reported only 50%.
- Klutke et al. (2001), from 600 retropubic MUS: 2.8% required reoperation for retention and persistent obstructive symptoms, performed at a mean of 64 days. The sling was either released with 1 cm of downward traction or cut in the midline. One urethral injury, repaired without sequelae. Symptoms resolved in ALL patients, all voided to completion, and nearly all remained continent.
- Most authors recommend waiting AT LEAST 2 WEEKS before sling incision for urethral obstruction; although one series reported return to normal voiding after 6 weeks of conservative management, the majority of studies do NOT support waiting beyond 4 weeks.
- Pinsard et al. (2023), comparing early loosening with delayed section or excision in patients with PVR ≥ 100 mL: complication rates were similar and withdrawal from CIC at 3 months was not significantly different, but at median 9 months significantly MORE patients had resolved voiding dysfunction in the EARLY group (63.2% vs 31.3%, $p = 0.01$), and persistent or recurrent SUI was HIGHER in the DELAYED group (21% vs 43.7%, $p = 0.04$). On multivariate analysis, the main predictive factor for recurrent SUI was DELAYED SECTION.
- Author's position: the sling should be incised WITHIN 4 WEEKS in cases of significant obstruction or retention. Author's clinical observation: patients who return to "normal voiding" with conservative management often retain persistent urgency and frequency.

Storage Symptoms After Mid-Urethral Sling

- Storage symptoms — urgency, frequency, UUI — are common after MUS, either as persistent or worsened symptoms in preoperative MUI, or de novo after surgery for pure SUI.
- Pergialiotis et al. (2017), systematic review of 32 studies and 3,139 patients across all three MUS types: de novo OAB in 11.5% of non-randomised studies and 6.4% of randomised studies. By device the rates are strikingly similar: SIMS 9.7%, inside-out transobturator 11.2%, outside-in transobturator 8.7%, TVT 9.8% — no route protects against de novo OAB.
- A broader literature review estimates de novo OAB syndrome at 15–30% (Marcelissen and Van Kerrebroeck, 2018).

- **Predictive factors:** older age, and urodynamic signs of OAB — detrusor overactivity, low bladder capacity, and elevated detrusor pressures.
- **Storage LUTS may develop or worsen without BOO — BUT a strong temporal relationship between the MUS and symptom onset or worsening should place PARTIAL BOO AT THE TOP OF THE DIFFERENTIAL.**
- **Storage symptoms and pain become MORE prevalent over weeks to months as the bladder attempts to adjust to the obstruction — a worsening trajectory suggests obstruction rather than idiopathic OAB.**
- **Optimal evaluation is poorly defined. Beyond detailed history and pelvic examination, CYSTOSCOPY IS MANDATORY.** Urodynamics are useful in selected cases at the physician's discretion.

Treatment

- The same as for the bladder neck sling. Conservative first: behavioural modification, PFMT, biofeedback, and pharmacotherapy with antimuscarinics or β_3 -adrenergic agonists.
- For persistent bothersome symptoms: decide between SLING REVISION and third-tier OAB therapy — sacral or posterior tibial nerve stimulation, or onabotulinumtoxinA.
- Miotla et al. (2016): onabotulinumtoxinA for de novo OAB after MUS was comparable in efficacy to treatment of idiopathic OAB — PROVIDED TAPE MALPOSITION WAS EXCLUDED. Complete dryness at 12 weeks in about two-fifths, with significant improvement in all secondary outcomes.
- Serati et al. (2014): patients with de novo OAB had significantly WORSE subjective and objective results after 3 months of daily solifenacin 5 mg than those with idiopathic OAB — antimuscarinics perform less well in post-sling OAB.

KEY POINTS: MID-URETHRAL SLING STORAGE AND EMPTYING DYSFUNCTION

- Long-term urinary retention and obstructive voiding dysfunction are both RARE after MUS.
- Definitions of retention and indications for management vary, but MOST patients may be managed conservatively initially.
- URETHRAL DILATION IS POTENTIALLY HARMFUL AND SHOULD NOT BE PERFORMED.
- For persistently elevated PVR with bothersome symptoms refractory to conservative management, TRANSVAGINAL SLING RELEASE consistently resolves symptoms while MAINTAINING CONTINENCE in the majority. Recommended waiting period: at least 2 to 4 weeks.
- Sling release should be attempted through a SMALL MIDLINE VAGINAL INCISION using MINIMAL DISSECTION. Patients obstructed longer than 3 months may require more extensive excision.
- Worsened or de novo storage LUTS should raise STRONG SUSPICION OF PARTIAL BOO. Treatment mirrors that for the autologous BNS, but outcomes of pharmacological and surgical treatment of de novo storage LUTS are limited.

SEXUAL DYSFUNCTION AFTER MID-URETHRAL SLING

- The relationship is INCOMPLETELY UNDERSTOOD, with varying degrees of sexual impairment reported.

Evidence of harm

- Mazouni et al. (2004), prospectively after retropubic MUS in previously unimpaired women: de novo dyspareunia in about a seventh, and decreased libido in a smaller minority.
- Stav et al. (2010b), in a much larger cohort of 1,112 patients: de novo dyspareunia in only 3% — the larger denominator gives the lower figure.
- Marszalek et al. (2007): deterioration in sexual function was significantly associated with de novo urgency, dyspareunia, and the sensation of incomplete emptying — i.e. it is the LUTS, not the mesh per se, that impairs sexual function.

- **Demirkesen et al. (2008)**, MUS versus Burch colposuspension: **about half expressed a negative change**, mostly dyspareunia; MUS appeared to affect satisfaction more adversely, but **NOT** significantly.
- **Kuhn et al. (2009)**: sling **REMOVAL** significantly improved de novo dyspareunia.

Evidence of benefit

- **Ward and Hilton (2008)**, in their randomised trial: dyspareunia **DECREASED** from 34% preoperatively to 13% at 5 years after sling surgery.
- **Bekker et al. (2009)**, in 136 patients after MUS: **improved sexual intercourse in about a fifth**, worsened in a small minority — the improvement attributed mainly to a significant decrease in **COITAL INCONTINENCE**.

ADJUSTABLE SLINGS

The Remeex readjustable sling — device description

- A short suburethral polypropylene mesh sling with non-absorbable suspension sutures attached to a tensioning device called a **VARITENSOR**.
- The mesh is placed under the mid-urethra as in a retropubic MUS.
- The varitensor is a **SCREW** attached to a **ROTATING REEL**, implanted **ABOVE THE RECTUS FASCIA** through a small suprapubic incision.
- Postoperatively the screw is accessed **PERCUTANEOUSLY** with a **MANIPULATOR** functioning like a screwdriver; turning it tightens suburethral sling tension, allowing **INDIVIDUALISED** tensioning.
- The manipulator is then withdrawn, leaving the varitensor in the subcutaneous tissue over the fascia and under the skin.
- **Proposed indication: COMPLEX SUI — ISD, fixed or hypomobile urethra, and reoperations** — precisely the population in which conventional MUS performs worst.

Outcomes

- **Giberti et al. (2011)**, in predominantly ISD patients at 5 years mean follow-up, cure defined as dry on cough stress test with pad weight of 0–1 g: **cure in about 86%**, with substantial reduction in mean pad weight. **Readjustment of tension was needed in a small minority**, and persistent retention, seroma and de novo urgency each occurred infrequently.
- **Giberti et al. (2017)** at 7 years mean follow-up: **cure in 90%**, with readjustment again needed in a small minority; **main adverse events were bacterial cystitis, de novo urgency, persistent retention and seroma**.
- **Ko et al. (2017)**, in patients with SUI and concomitant DETRUSOR UNDERACTIVITY: **treatment success in about four-fifths — but despite being dry, a quarter wanted the Remeex system REMOVED because of persistent postoperative retention within a year**. Dryness is not the same as satisfaction in this group.
- **Ros et al. (2022)**, in 120 females: **subjective success 70%**, defined as improvement by one or more severity levels on the International Consultation on Incontinence Questionnaire — Urinary Incontinence Short Form (ICIQ-UI-SF). **Greater BMI and greater presurgical 24-hour pad weight predicted WORSE results**.
- **Kim et al. (2019c)**, by contrast: **subjective success above 90% in normal-weight, overweight and obese groups alike, with high and statistically indistinguishable satisfaction**.

Complications and position

- **Overall complication rates 32–51%**, but **MOST** are Clavien class II and III.
- **Bladder injury 6.7–14%**. **De novo OAB 7.8–12.3%**. **Readjustment rates vary very widely, from about 14% to 43%**.

- **The adjustable MUS is an intriguing option for a challenging subset with few other effective options.**
- **Promising success must be balanced against the potential need for PERIODIC TENSION ADJUSTMENTS.**
- **LEVEL 1 EVIDENCE WOULD BE WELCOME** — none exists.